

Computer Networks



Course Code: CSC263
Year/ Semester: II/IV

Compiled by Shishir Ghimire

Credit Hours: 3hrs



Unit - 06: Application Layer

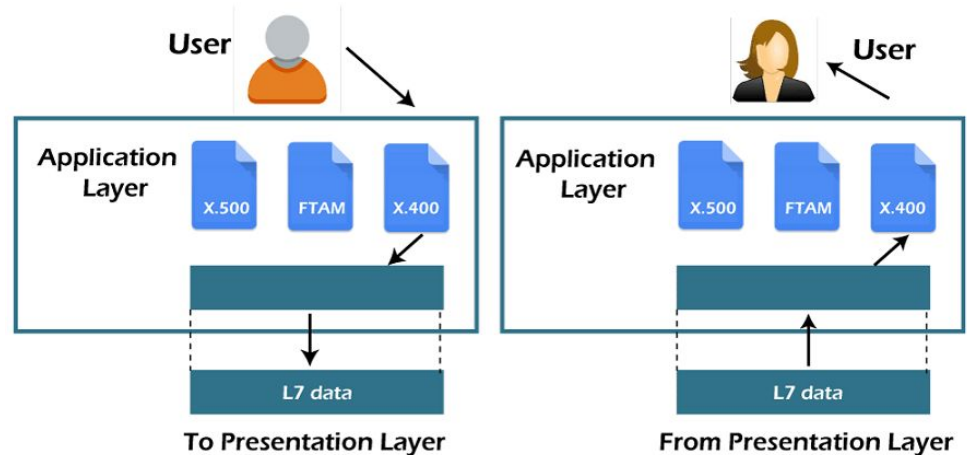
Class Load : 7 hrs

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► Application Layer:

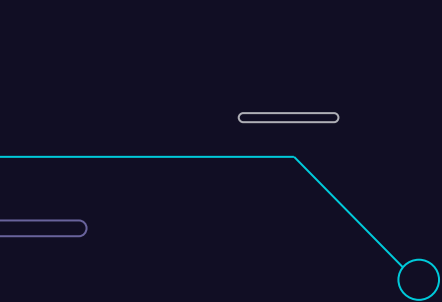
- Application layer allows **access to** network resources.
- In the application layer, an application, such as an internet browser, gets the data and a user can then interact with it.
- **Highest level** of OSI layers.
- Application-layer helps to **identify** communication partners, **determining** resource availability, and **synchronizing** communication.
- It allows users to **log on** to a remote host.
- This layer provides various e-mail services.
- This is also called **closest layer** from user.



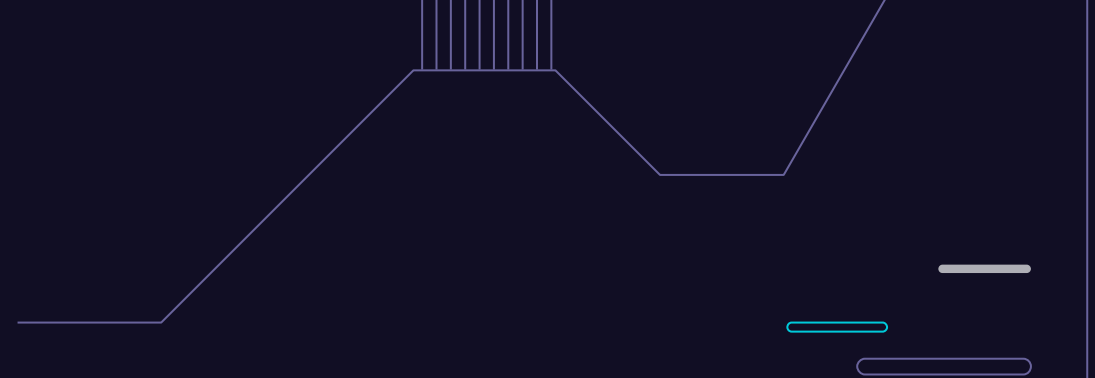
► Application Layer:

Functions of Application layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.
- **Directory services:** An application provides the distributed database sources and is used to provide that global information about various objects.



▶ Introduction & Functions



6.1

► Introduction:

- ❖ Application layer is where the **actual communication** is initiated.
- ❖ Application layer takes the **help of Transport layer** and all layers below it to communicate data to the **remote host**.
- ❖ Application layer provides the **interface between** the applications, used to communicate to underlying network.
- ❖ It is the layer **closest to the** end user, implying that both the application layer and the end user can **interact** with the software application directly.
- ❖ Users can **send** data, **access** data, and **use** networks through the application layer.
- ❖ The layers also make communication easier and, in some cases, allow users to use software programs. Although the software is **not part** of the OSI model, the application layer allows users to access programs and information.

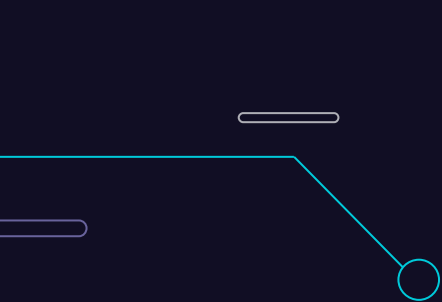
► Introduction:

- ❖ The application layer protocols makes **communication between** sender and receiver faster, more **efficient, reliable**, and **secure**.
- ❖ These protocols will be discussed further below.
 - WEB
 - HTTP
 - HTTPs
 - SMTP
 - IMAP
 - POP
 - FTP
 - DNS
 - MIME-Multipurpose Internet Mail Extension

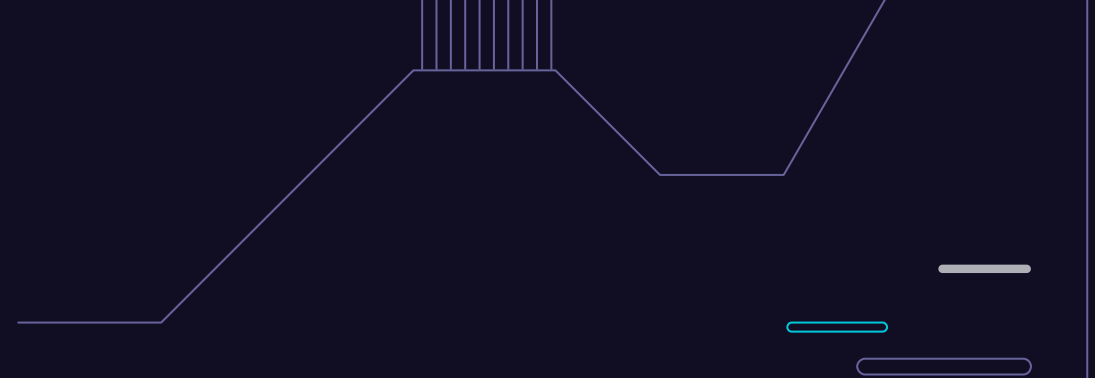
► Application Layer:

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► Web & HTTP

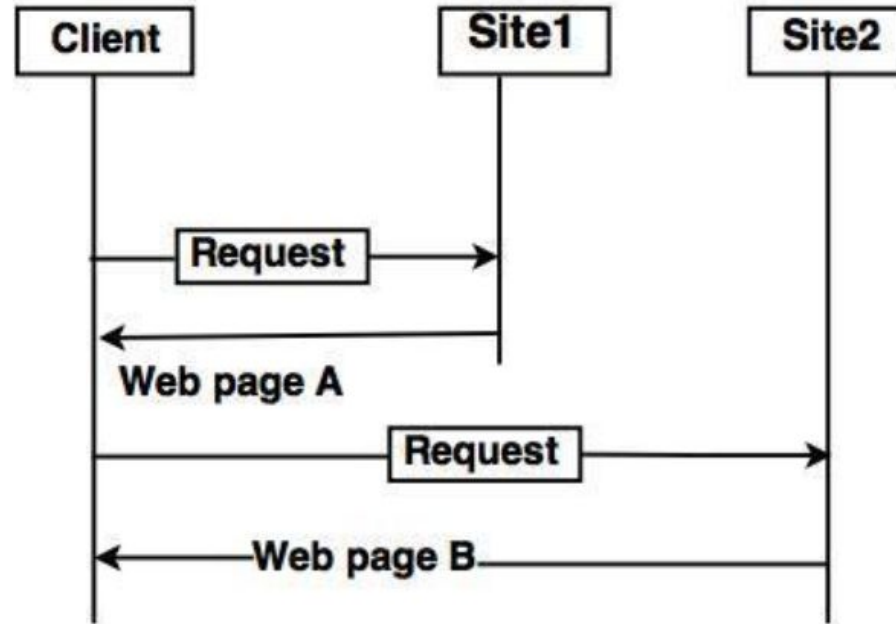


6.2

► Overview:

- ❖ The World Wide Web (WWW) is a **collection** of documents and other web resources which are **identified by URLs**, **interlinked** by hypertext links, and can be **accessed and searched** by browsers via the Internet.
- ❖ World Wide Web is also called the Web and it was **invented by** Tim Berners-Lee in 1989.
- ❖ Website is a collection of web pages belonging to a **particular organization**.
- ❖ It is a **client-server service**, in which a client using a browser can access a service through server.
- ❖ The service is distributed over many locations called **websites**.
- ❖ The WWW uses the concept of **hypertext** and **hypermedia**.
- ❖ Information is stored in a set of documents that are linked together using **hyperlinks**.
- ❖ To use WWW, **three components** are needed: a **browser**, a **web server** and a **protocol** called Hypertext Transfer Protocol(HTTP).

► WWW Architecture:



Architecture of WWW

► Browser:

- ❖ A browser is an **application program** that provides a way to look at and interact with all the information on the World Wide Web.
- ❖ Browsers **interpret** and **display** a web document.
- ❖ Browsers consists of **three parts**: a controller, client programs and interpreters.
 - **Controller**: Controller **receives** input from the keyboard or the mouse and uses the **client programs** to **access** the document.
 - **Client Program**: Client program can be **one of the protocols** : FTP or HTTP
 - **Interpreter**: After the document is accessed, the controller uses one of the **interpreters** to display the document **on screen**.

► HTTP (Hypertext Transfer Protocol):

- ❖ HTTP refers to "Hypertext Transfer Protocol."
- ❖ It is a protocol used to **access data** on the World Wide Web (WWW).
- ❖ The protocol transfers data in the form of plain text, hypertext, audio, video and so on.
- ❖ HTTP is the protocol that can transfer information **over the network**.
- ❖ It is the Internet protocol suite method and defines commands and functions used for **sharing web** page data.
- ❖ **HTTPS** is the more secured version of HTTP, that's why HTTPS stands for Hypertext Transfer Protocol Secure.
- ❖ This protocol is used to access data from the World Wide Web.
- ❖ HTTP is based on the **client-server model**.
- ❖ It uses **TCP** for establishing connections.

► HTTP (Hypertext Transfer Protocol):

- ❖ HTTP is a **stateless protocol**, which means the server doesn't maintain any information about the previous request from the client.
- ❖ HTTP uses **port number 80** for establishing the connection.
- ❖ HTTP uses a **client-server** model. A client, for example, maybe a laptop or telephone device.
- ❖ The HTTP server is frequently a web host running web server software, such as **Apache** or IIS.
- ❖ HTTP also represents commands such as **GET and POST**, which are used to handle submissions on websites.
- ❖ The **CONNECT** command can act as a **fast connection** that is encrypted using a **secure socket layer (SSL)**.

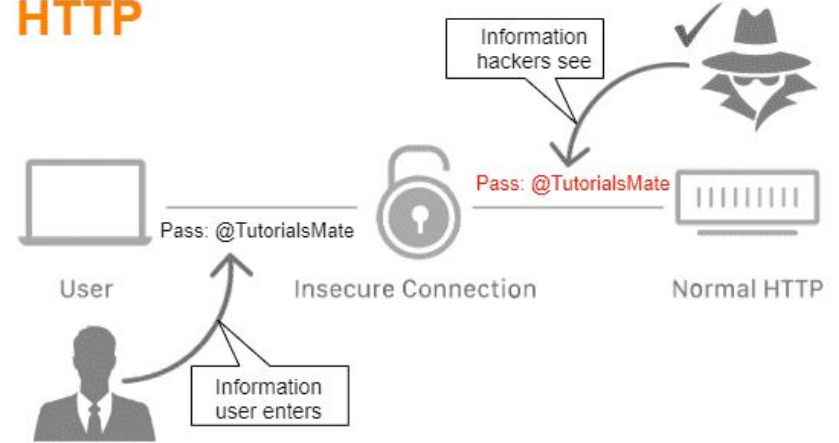
► HTTPS (HyperText Transfer Protocol Secure):

- ❖ Hypertext Transfer Protocol Secure (HTTPS) is the **secure version of HTTP**, the protocol over which data is sent between your browser and the website that you are connected to.
- ❖ The 'S' at the end of HTTPS stands for 'Secure'. It means all communications between your browser and the website are **encrypted**.
- ❖ HTTPS is often **used to protect** highly confidential online transactions like online banking and online shopping order forms.
- ❖ HTTPS pages typically use one of **two secure protocols** to encrypt communications - **SSL** (Secure Sockets Layer) or **TLS** (Transport Layer Security).
- ❖ Both the TLS and SSL protocols use what is known as an **'asymmetric'** Public Key Infrastructure (PKI) system. An asymmetric system uses **two 'keys'** to encrypt communications, a **'public' key** and a **'private' key**.

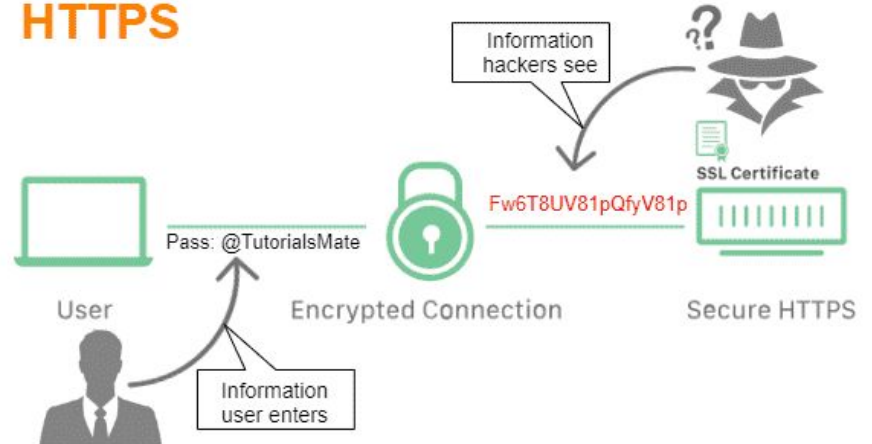
► HTTP vs HTTPS



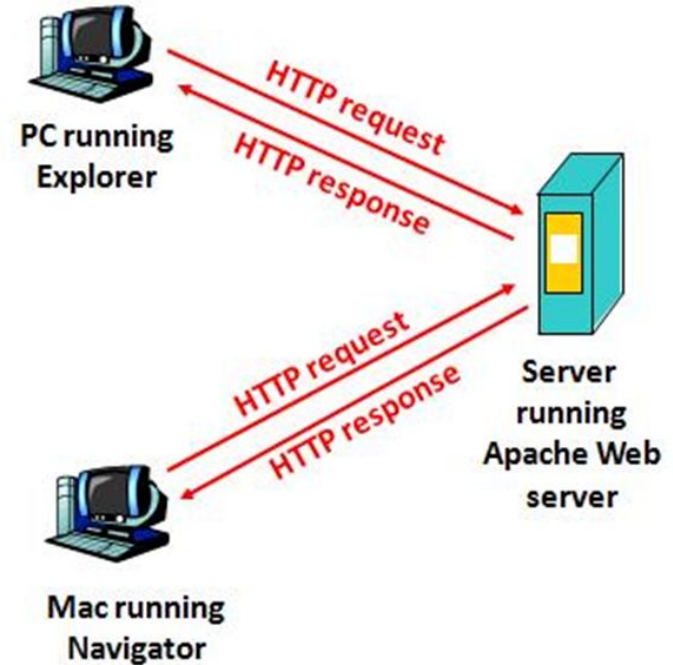
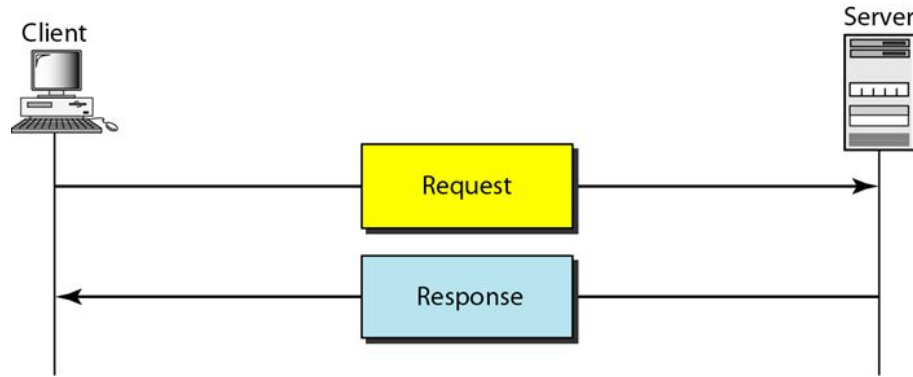
HTTP



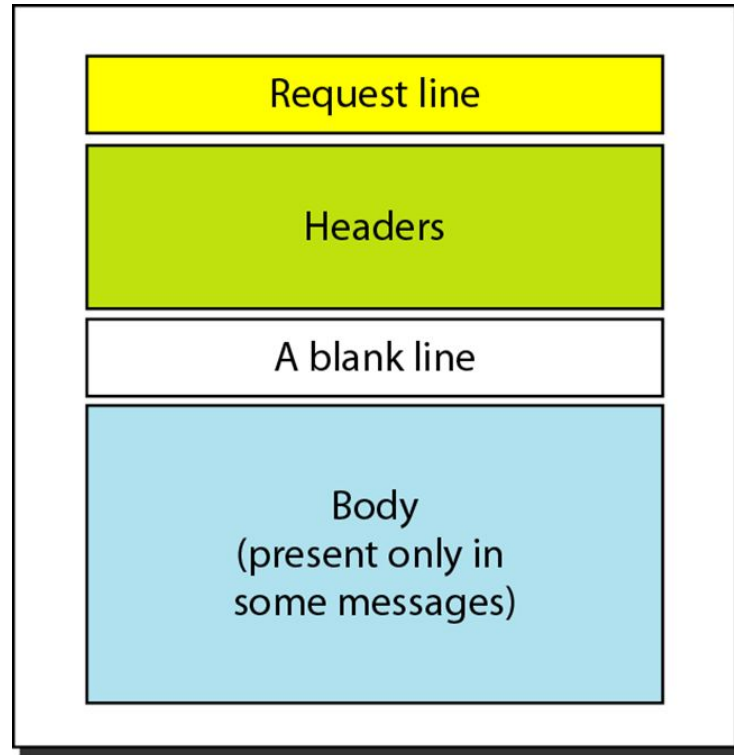
HTTPS



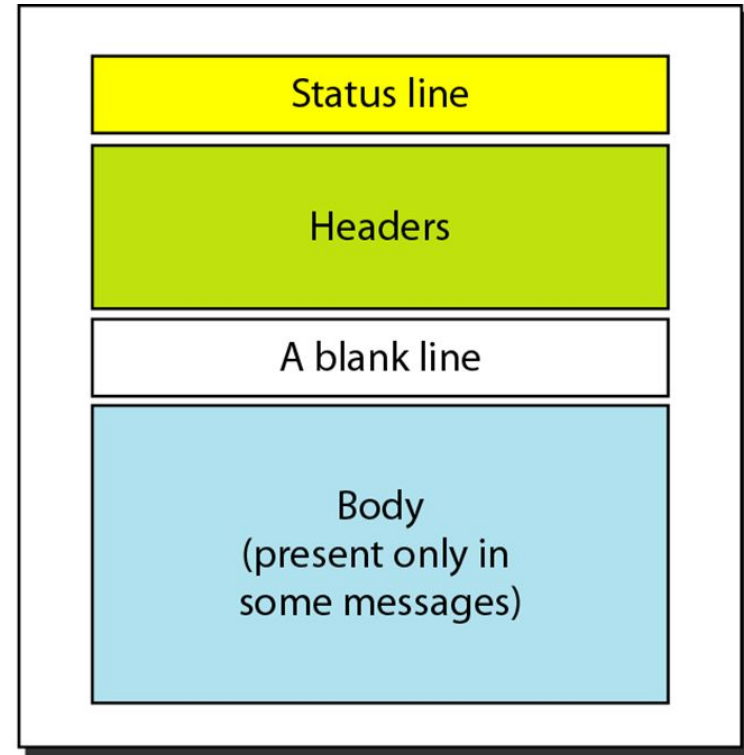
► HTTP (Request/ Response Protocol):



► HTTP (Request/ Response Protocol):



Request message



Response message

► HTTP (Request/ Response Protocol):

Request

Response

```
POST / HTTP/1.1
Host: developer.mozilla.org
User-Agent: curl/8.6.0
Accept: */*
Content-Type: application/json
Content-Length: 345

{
  "data": "ABC123"
}
```

← Start line →

← Headers →

← Empty line →

← Body →

```
HTTP/1.1 403 Forbidden
Server: Apache
Date: Fri, 21 Jun 2024 12:52:39 GMT
Content-Length: 678
Content-Type: text/html
Cache-Control: no-store

<!DOCTYPE html>
<html lang="en">
  (more data...)
```

This first line (`POST/hello.htm`
`HTTP/1.1`) is called the **Request-Line**.

This first line (`HTTP/1.1 403 Forbidden`)
is called the **Status-Line**.

► HTTP : Request and Response Messages Methods:

<i>Method</i>	<i>Action</i>
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
POST	Sends some information from the client to the server
PUT	Sends a document from the server to the client
TRACE	Echoes the incoming request
CONNECT	Reserved
OPTION	Inquires about available options

► HTTP Status Code:

- ❖ 1xx: Informational Response
- ❖ 2xx: Successful
- ❖ 3xx: Redirection
- ❖ 4xx: Client Error
- ❖ 5xx: Server Error

S.N.	Code and Description
1	1xx: Informational Response It is used to show that the request was received, and the process is continuing.
2	2xx: Successful It is used to show that the request was successfully received, understood, and accepted.
3	3xx: Redirection It is used to show that further action needs to be taken to complete the request.
4	4xx: Client Error It is used to show that the request contains bad syntax or cannot be fulfilled.
5	5xx: Server Error It is used to show that the server is failed to fulfill an apparently valid request.

► HTTP : Status Codes ► 1XX and 2XX

<i>Code</i>	<i>Phrase</i>	<i>Description</i>
Informational		
100	Continue	The initial part of the request has been received, and the client may continue with its request.
101	Switching	The server is complying with a client request to switch protocols defined in the upgrade header.
Success		
200	OK	The request is successful.
201	Created	A new URL is created.
202	Accepted	The request is accepted, but it is not immediately acted upon.
204	No content	There is no content in the body.

► Features of HTTP:

❖ Connectionless protocol:

- HTTP is a connectionless protocol. HTTP user **initiates a request** and waits for a **response from** the server.
- When the server gets the request, the server processes the request and sends back the response to the HTTP user, after which the **client disconnects** the connection.

❖ Media independent

- HTTP protocol is media independent as data can be transmitted **as long as both** the user and server know **how to manage** the data content.

► Features of HTTP:

◆ Stateless:

- It is a stateless protocol as both the client and server learn each other **only during the current request.**
- In HTTP every client connection **opens a new session** that sends its request the stateless nature keeps the protocol very simple and straightforward.
- This consumes very few resources on the server and can support more simultaneous users since there are no client information **overheads to be maintained** throughout the sessions.

► HTTP connections:

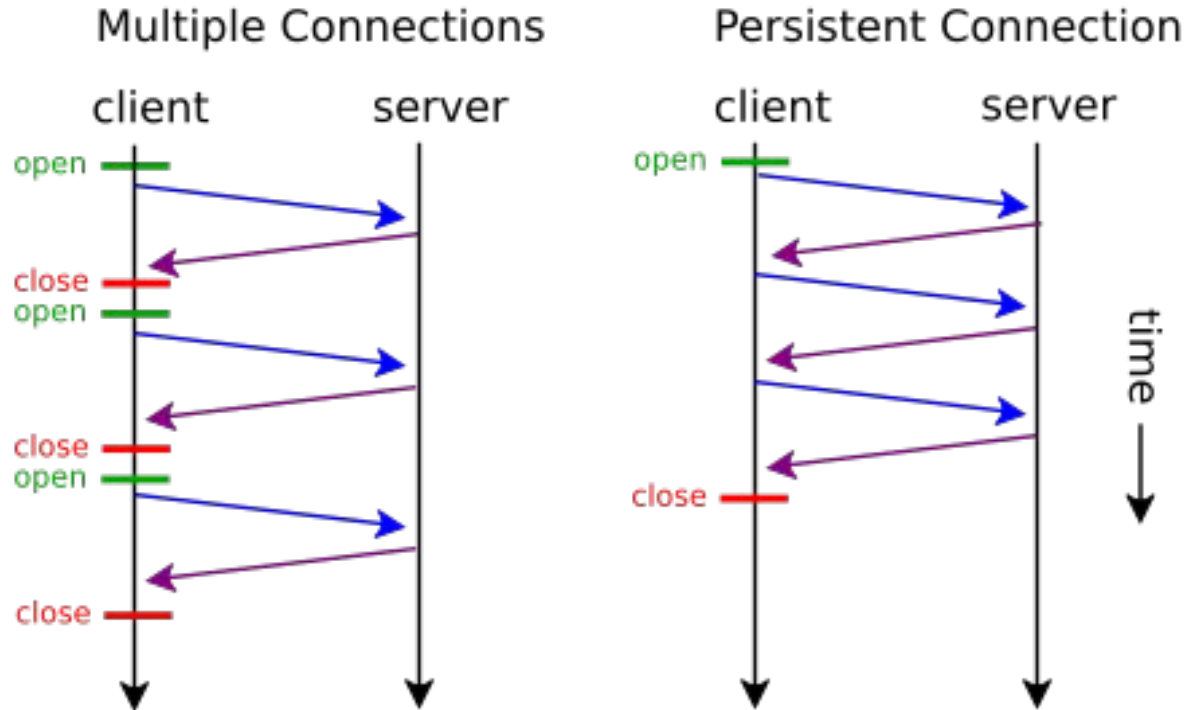
◆ Persistent Connection

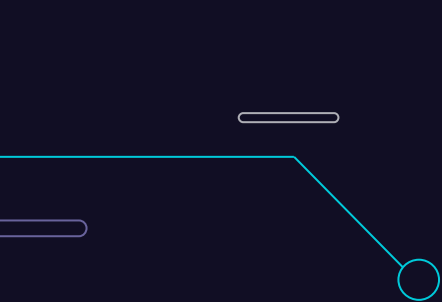
- **Multiple objects** can be sent over single TCP connection between client, server.

◆ Non-Persistent Connection

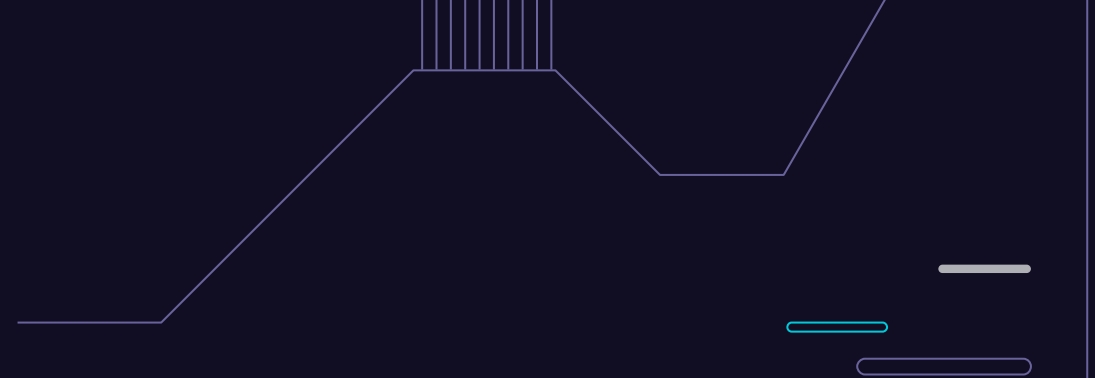
- **At most** one object sent over TCP connection
- connection **then closed**
- downloading multiple objects **requires multiple** connections

► Non-Persistent vs Persistent HTTP Connection:





► DNS and the Query Types



6.3

► DNS:

- ❖ DNS is a global system for **translating** IP addresses to human-readable domain names.
- ❖ When a user tries to access a web address like “example.com”, their web browser or application **performs a DNS Query** against a **DNS server**, supplying the hostname.
- ❖ The DNS server takes the hostname and resolves it into a **numeric IP address**, which the web browser can connect to.

◆ **DNS : Domain Name System**

- ❖ **Hierarchical** Naming System Built on a Distributed Database.
- ❖ Responsible to **Translate** Human Address into IP Address.
- ❖ DNS can use the services of UDP or TCP using the well-known **port 53**.

► DNS:

```
C:\Users\IC>nslookup www.google.com
Server:    UnKnown
Address:   192.168.42.129
```

```
Non-authoritative answer:
Name:      www.google.com
Addresses: 2404:6800:4005:80f::2004
           172.217.26.164
```

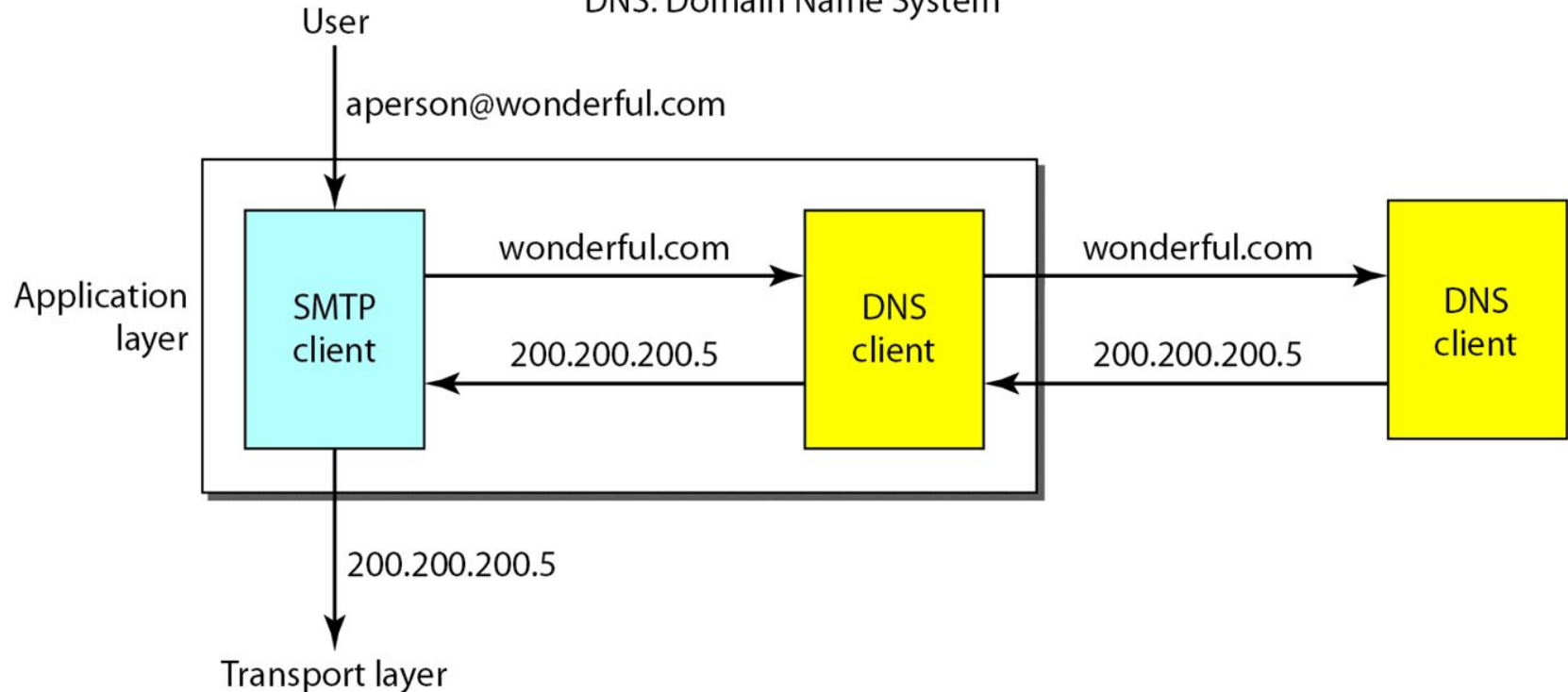
```
C:\Users\IC>nslookup www.facebook.com
Server:    UnKnown
Address:   192.168.42.129
```

```
Non-authoritative answer:
Name:      star-mini.c10r.facebook.com
Addresses: 2a03:2880:f10c:83:face:b00c:0:25de
           157.240.7.35
Aliases:   www.facebook.com
```

► DNS : Example of using the DNS service

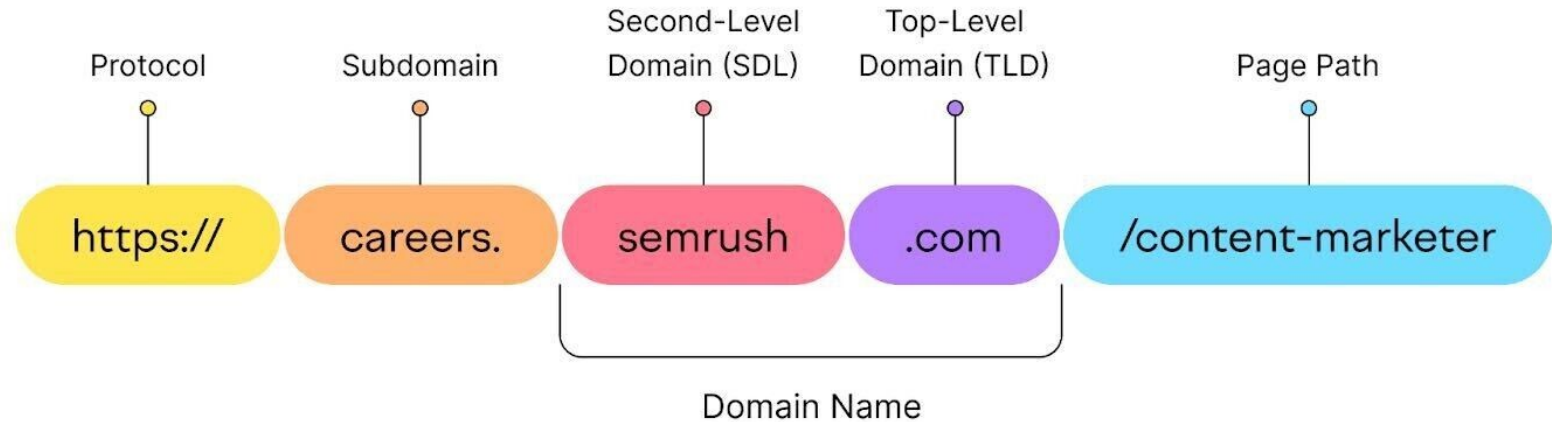
SMTP: Simple Mail Transfer Protocol (e-mail)

DNS: Domain Name System



► Parts of URLs:

Parts of a URL



► Types of DNS Server:

1. DNS Recursive Resolver/DNS Resolver
2. Root Name Server
3. Top Level Domain/ TLD Name Server
4. Authoritative Name Server

1. DNS Recursive Resolver/DNS Resolver

- A DNS resolver (recursive resolver), is designed to **receive DNS queries**, which include a human-readable hostname such as “www.example.com”, and is responsible for **tracking the IP address** for that **hostname**.

► 2.Root Name Server

- ❖ The root server is the **first step** in the journey from hostname to IP address.
- ❖ The DNS Root Server extracts the **Top Level Domain (TLD)** from the user's query. For example, www.example.com —and provides **details for the .com** TLD Name Server.
- ❖ In turn, that server will provide details for domains with the .com DNS zone, including "example.com".
- ❖ There are **13 root servers** worldwide, indicated by the letters **A through M**.

▶ 3.TLD(Top Level Domain)

- ❖ The TLD server holds records for **where to find the authoritative server** to answer the DNS request.

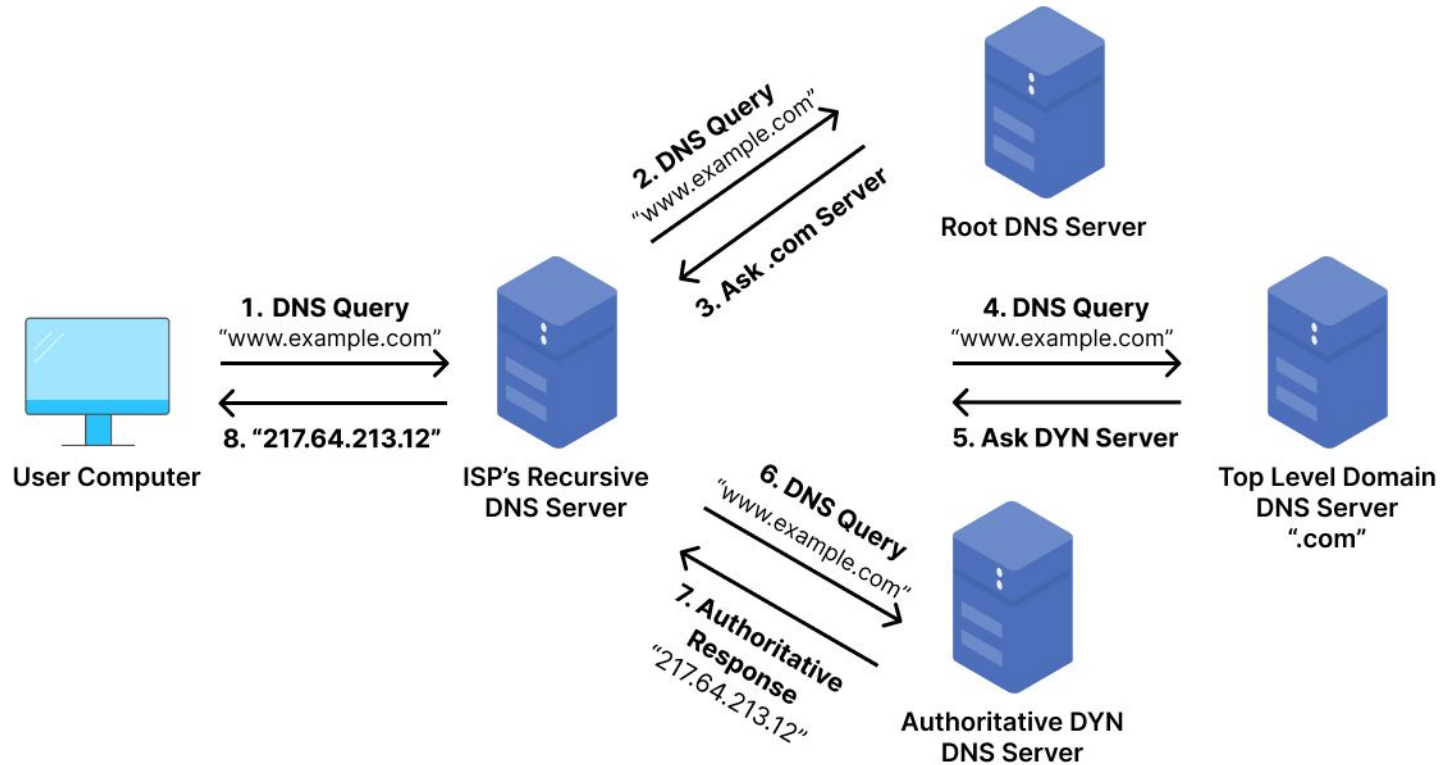
Examples of Top-Level Domains

Traditional gTLDs	New gTLDs	ccTLDs
.com .net .org .gov .edu .info .biz ...	.guru .plumbing .realtor .space .design .cloud .ninja ...	.ca .tw .uk .fr .de .in .br ...

► 4. Authoritative Name Server

- ❖ The authoritative server is often also known as the **nameserver** for the domain.
- ❖ Higher level servers in the DNS hierarchy define which **DNS server** is the “authoritative” name server for a **specific hostname**, meaning that it holds the **up-to-date information** for that hostname.
- ❖ The Authoritative Name Server is the **last stop** in the **name server query**: it takes the hostname and **returns the correct IP address** to the DNS Resolver (or if it cannot find the domain, returns the message “no domain”).

► How DNS Works?



► How DNS Works?

- ❖ DNS resolver is **provided by ISPs** which connects the **browsers** to the DNS Name Servers.
- ❖ When a user requests for a website, the request first goes to the **DNS resolver**.
- ❖ The **DNS resolver** then checks **its cache**, if it has the IP address of the website.
- ❖ **If not**, then it **forwards** the request to **Root Name Server**.
- ❖ The **Root Name Server** then checks the extension(TLDs) of the website. Whether it is .com, .org, etc.
- ❖ Then based on the extension, Root Name Server provides the IP address of **TLD Name Server** to the DNS resolver.
- ❖ DNS resolver then contacts the **respective TLD name server**.

► How DNS Works?

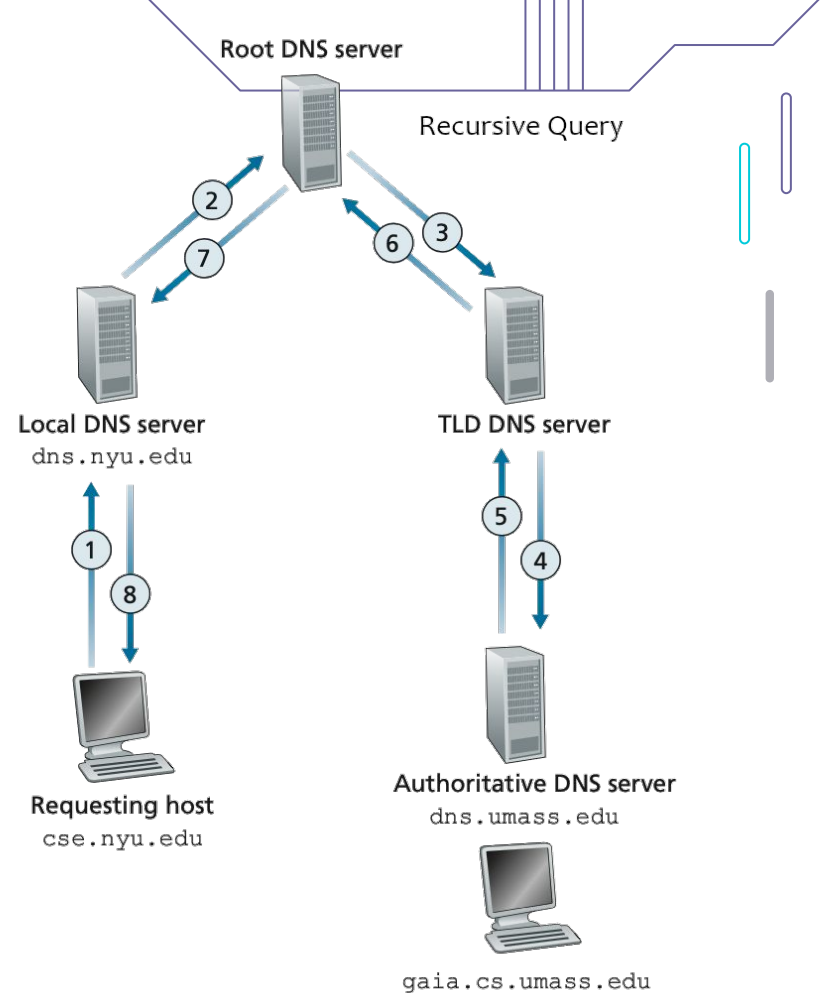
- ❖ **TLD name server** then provides the IP address of **Authoritative Name Server** which stores the IP address of requested website to DNS resolver.
- ❖ **DNS resolver** stores this **IP address** for future use and provides it to the requesting **browser**.
- ❖ **Note:**
 - When hosting a website, **during domain registration**, Authoritative name server contacts **TLD name server** to update about the **new added domain**.
 - Thus, TLD name server knows about the specific Authoritative name server IP address for specific website.

► DNS Query Types:

❖ Recursive Query

- In a recursive query, a DNS client provides a hostname and the DNS resolver must provide an answer.
- It responds with **either** a relevant resource record, **or** an error message if it can't be found i.e. gives definitive answer.
- Can't refer the DNS client to **another** DNS Server.
- The resolver starts a **recursive query process**, starting from the DNS Root Server, **until it finds** the Authoritative Name Server.
- Query made by DNS client to a DNS server.

► DNS: Name Resolution: Recursive Query

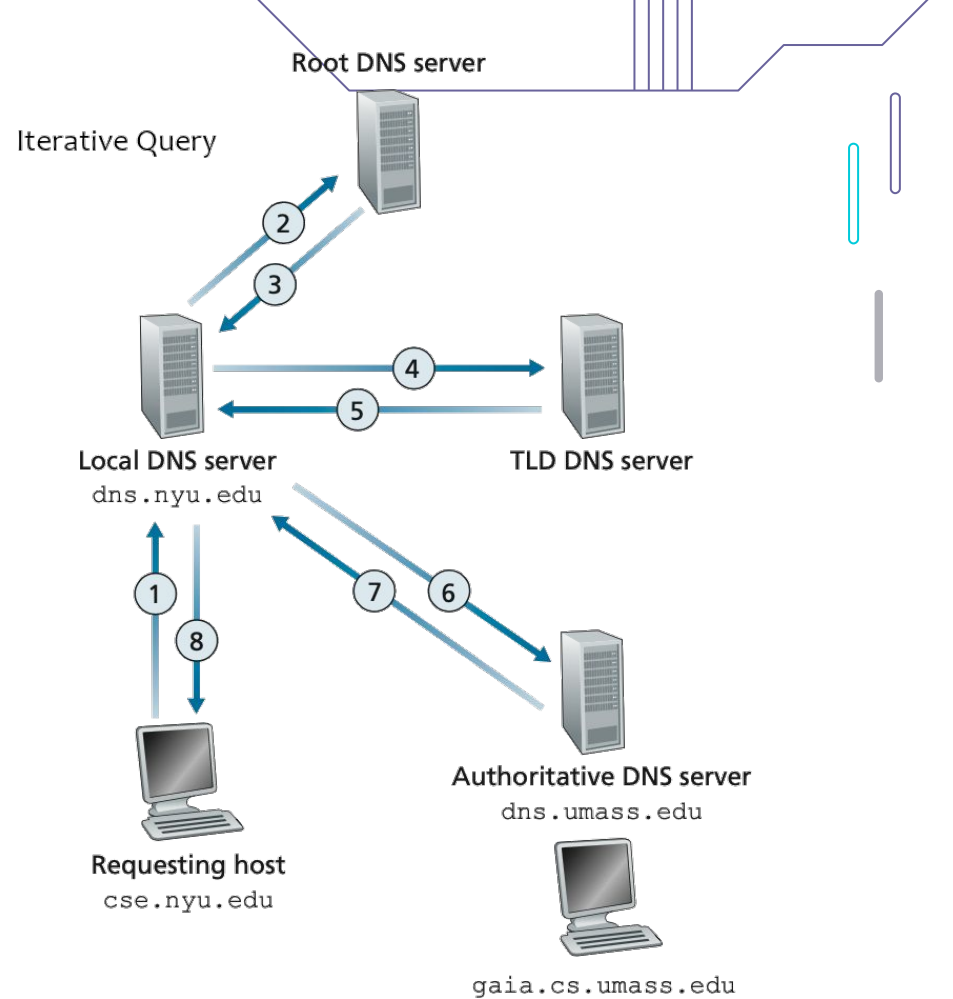


► DNS Query Types:

❖ Iterative Query

- Here, a DNS client provides a **hostname**, and the DNS Resolver **returns** the best answer it can.
- If the DNS resolver has the **relevant DNS records** in its cache, it returns them.
- **If not**, it refers the DNS client to the Root Server, or **another** Authoritative Name Server which is **nearest** to the required DNS zone.
- The DNS client must then repeat the **query directly** against the DNS server it was referred to.

► DNS Name Resolution: Iterative Query



► DNS Query Types:

❖ Non-Recursive Query

- A non-recursive query is a query in which the DNS Resolver **already knows** the answer.
- It either **immediately returns** a DNS record because it **already stores** it in local cache, or **queries** a DNS Name Server which is authoritative for the record, meaning it definitely holds the **correct IP for that hostname**.
- In both cases, there is **no need for additional rounds of queries** like in recursive or iterative queries.

► Non-Recursive Query:



Recursive Query

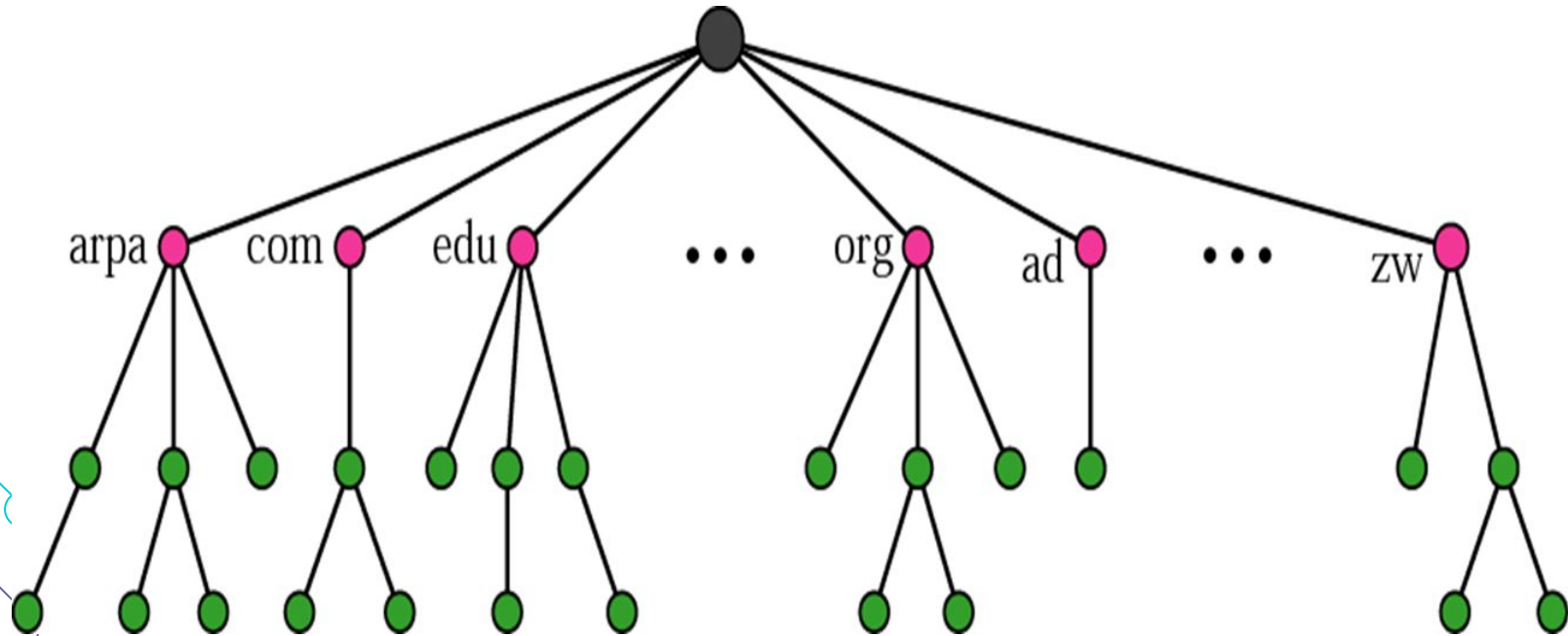


Non Recursive Query

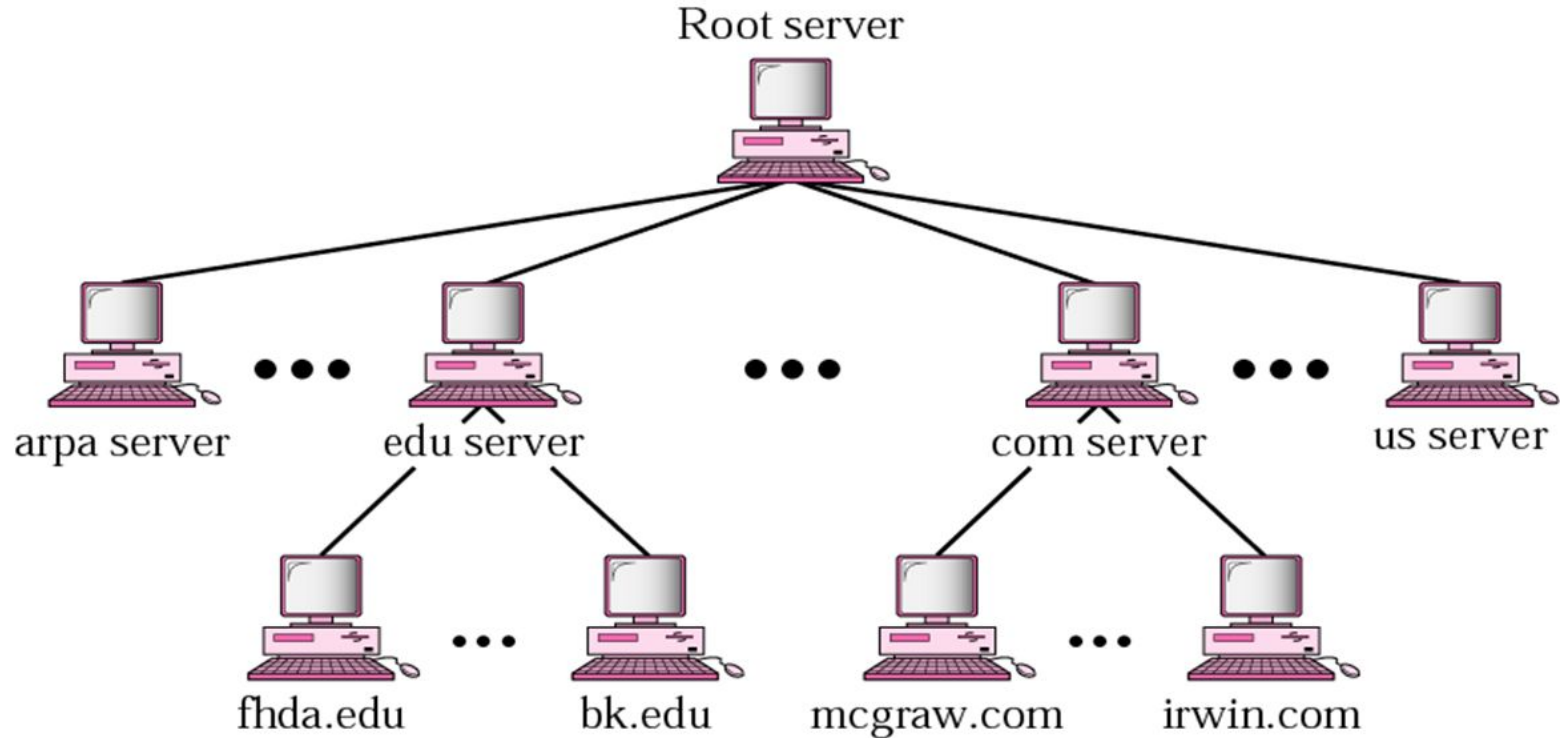
► DNS Name Space and Domain Name Space

- ❖ To be unambiguous, the names assigned to machines must be carefully selected from a namespace with **complete control over the binding** between the names and IP addresses
- ❖ **Two Types** of Namespace are:
 - Flat Name Space and
 - Hierarchical Name Space
- ❖ To have a hierarchical name space, a domain name space was designed.
- ❖ In this design the names are defined in an **inverted-tree structure** with the root at the top.
- ❖ The tree can have only 128 levels: level 0 (root) to level 127.

► DNS: Hierarchical Naming



► DNS: Hierarchy of Name Servers



► DNS: FQDN and PQDN

❖ FQDN Fully Qualified Domain Name

- A fully qualified domain name, also known as an **absolute domain name**, specifies all domain levels, **including** the top-level domain (TLD), in **hostname.domain.TLD** format.

❖ PQDN Partially Qualified Domain Name:

- A Partially Qualified Domain Name (PQDN) is a web address that **doesn't include** the **full path** specified by an FQDN. It might consist of only the hostname or a part of the domain structure, **excluding** the top-level domain (TLD).

FQDN

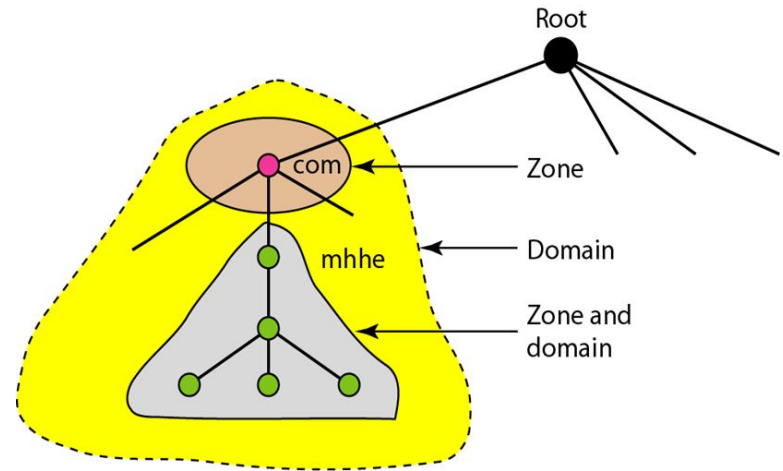
challenger.atc.fhda.edu.
cs.hmme.com.
www.funny.int.

PQDN

challenger.atc.fhda.edu
cs.hmme
www

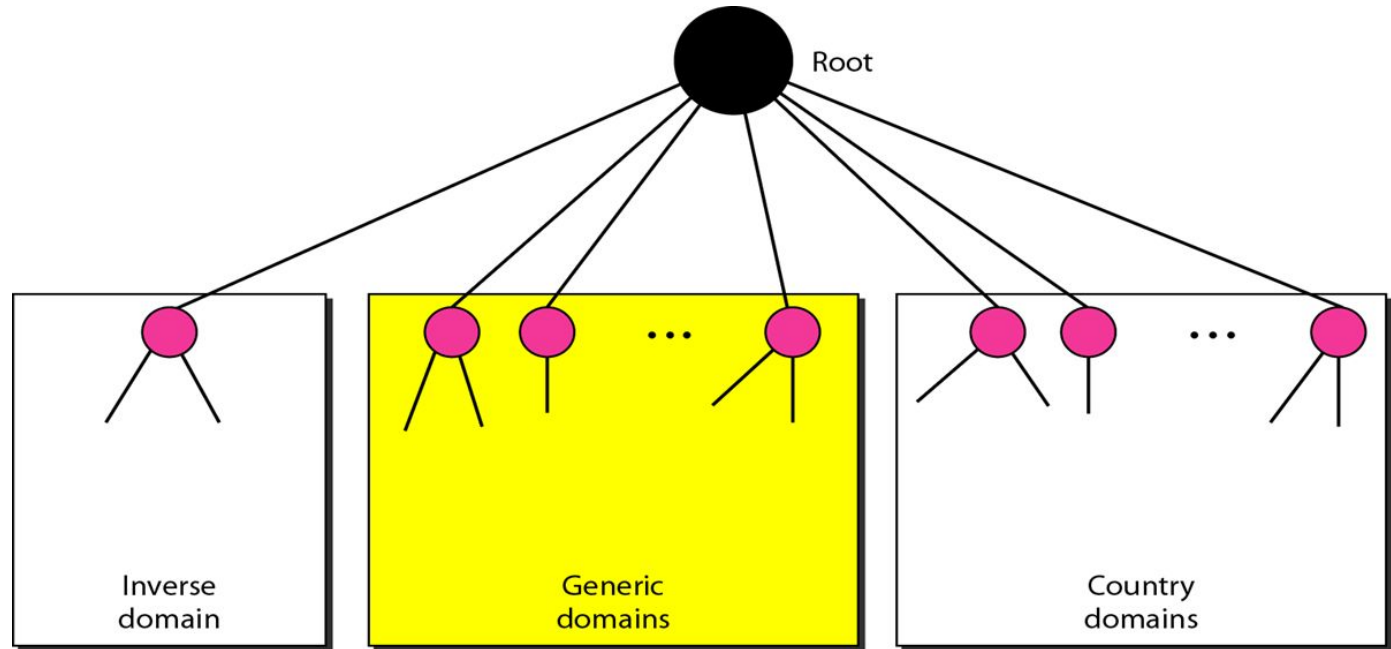
► DNS : Zones and domains

- ❖ A **primary server** loads **all information** from the disk file;
- ❖ The **secondary server** loads all information from the **primary server**.
- ❖ When the secondary downloads information from the primary, it is called **zone transfer**.

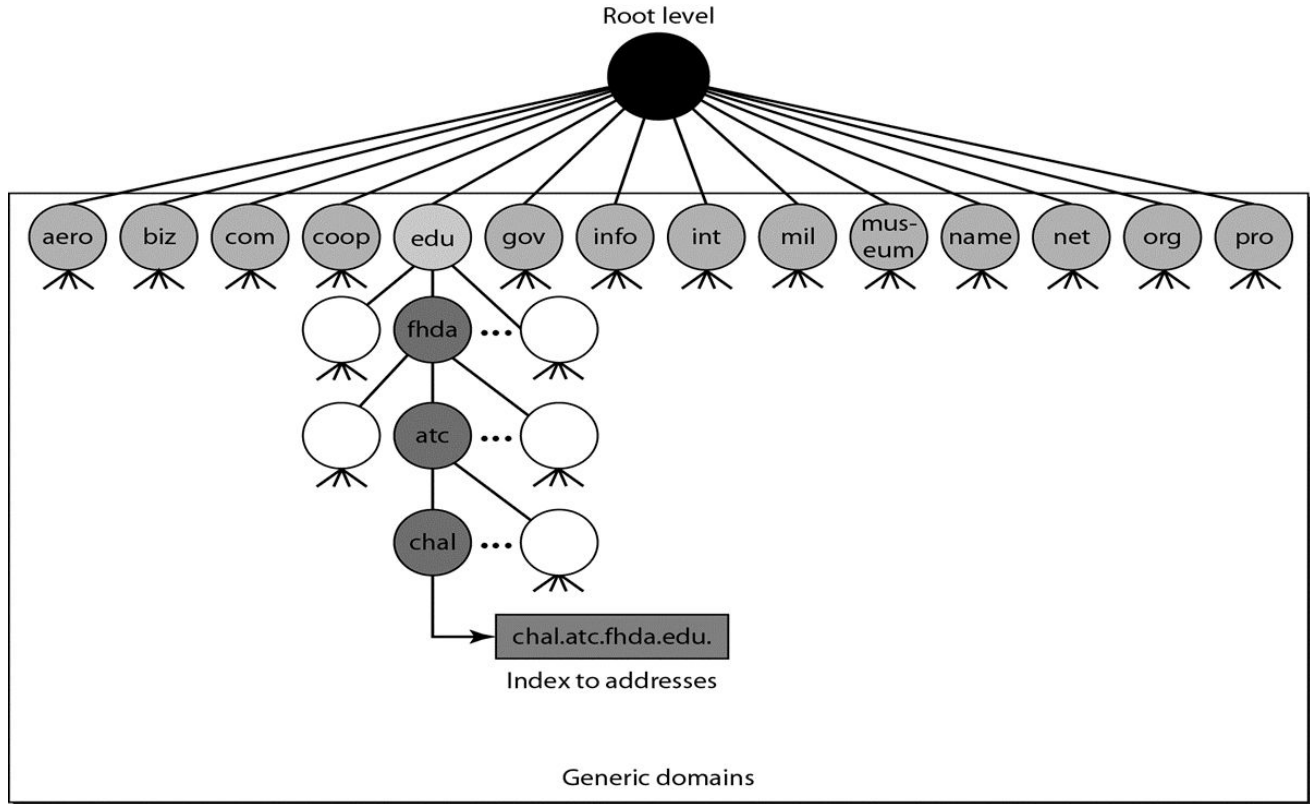


► DNS : DNS in the Internet

- ❖ In the Internet, the **domain name space (tree)** is divided into three different sections:



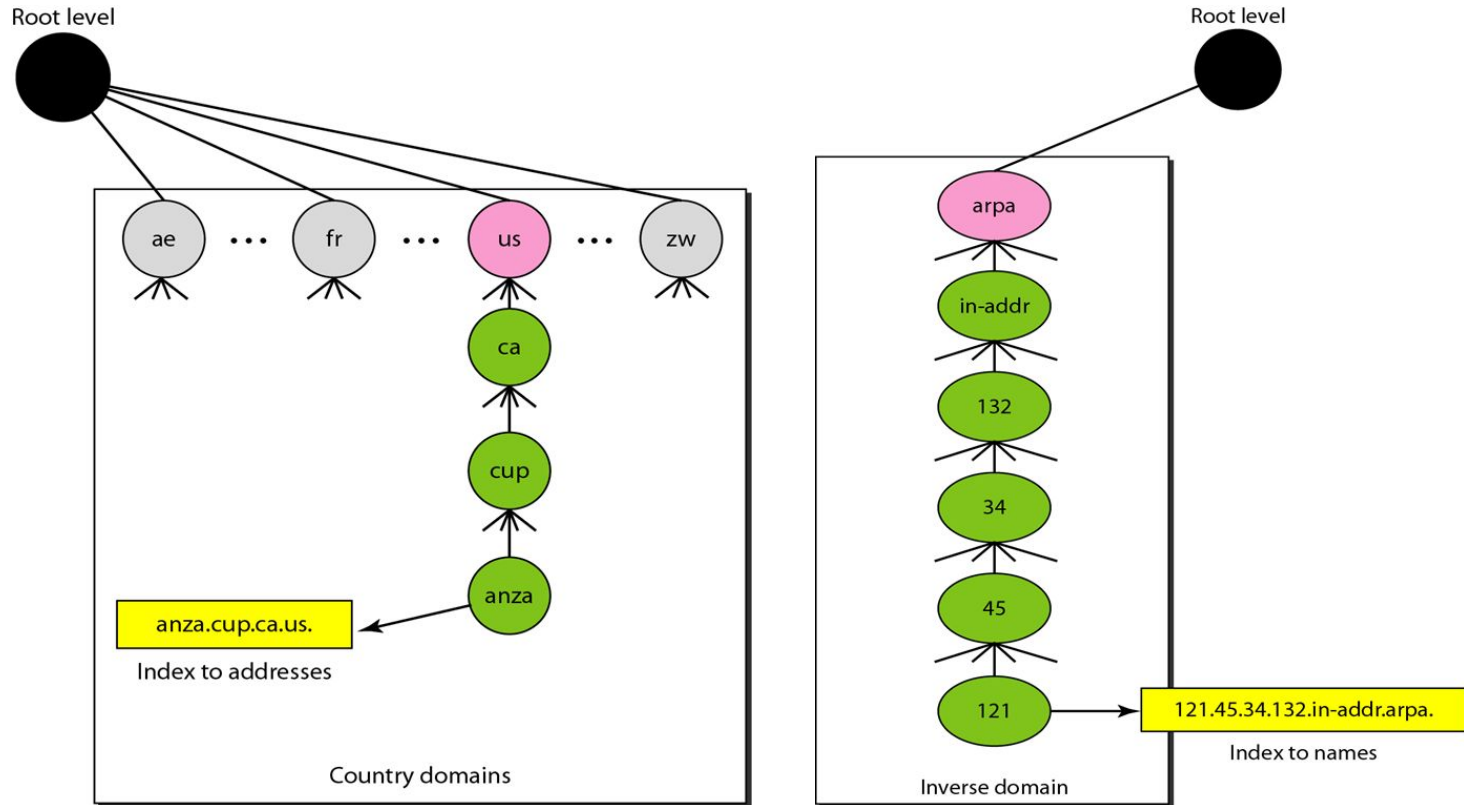
► DNS : Generic domain



► DNS : Generic domain labels

<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to “com”)
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

► DNS: Country domain and Inverse domain



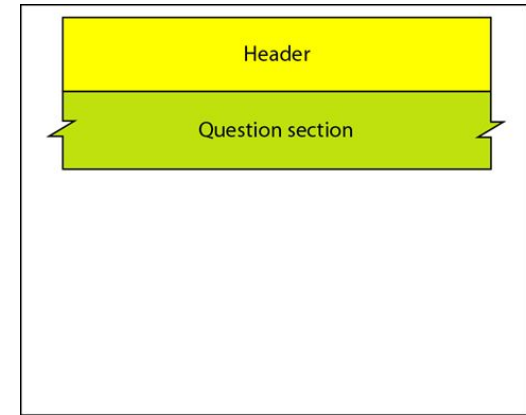
► DNS : Root Name Servers



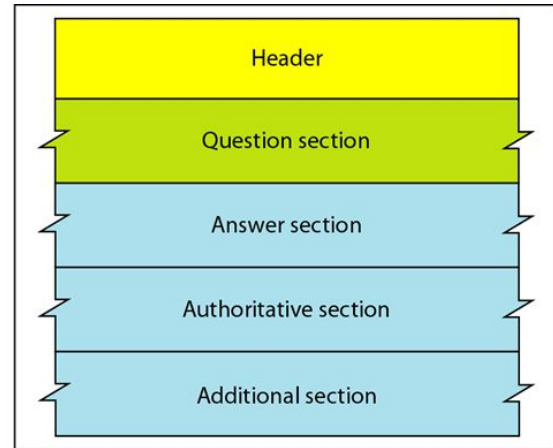
Figure: 13 Root Name Servers Worldwide

► DNS: Messages

- ❖ DNS has **two types** of messages:
 - query
 - response.
- ❖ Both types have the **same** format.
- ❖ The **query message** consists of
 - a header and
 - question records
- ❖ The **response message** consists of
 - a header,
 - question records,
 - answer records,
 - authoritative records, and
 - additional records.



a. Query

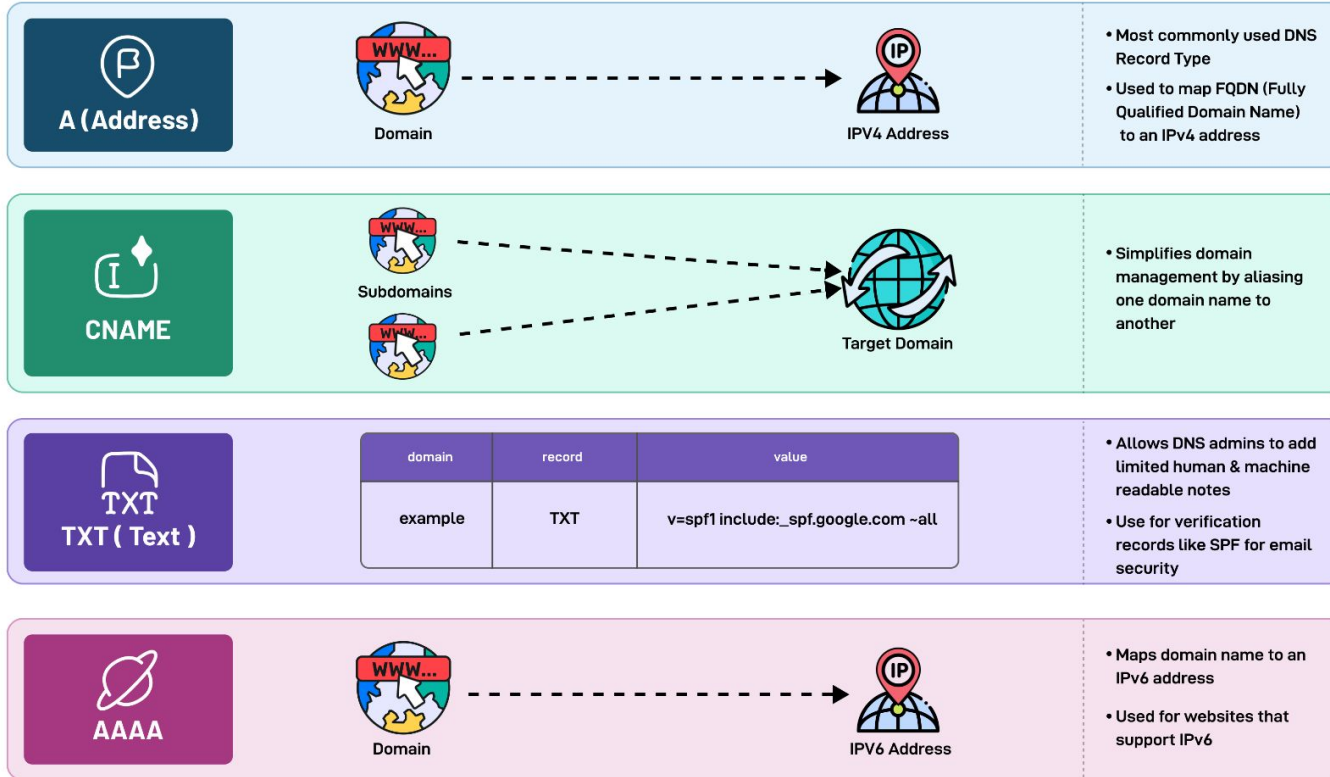


b. Response

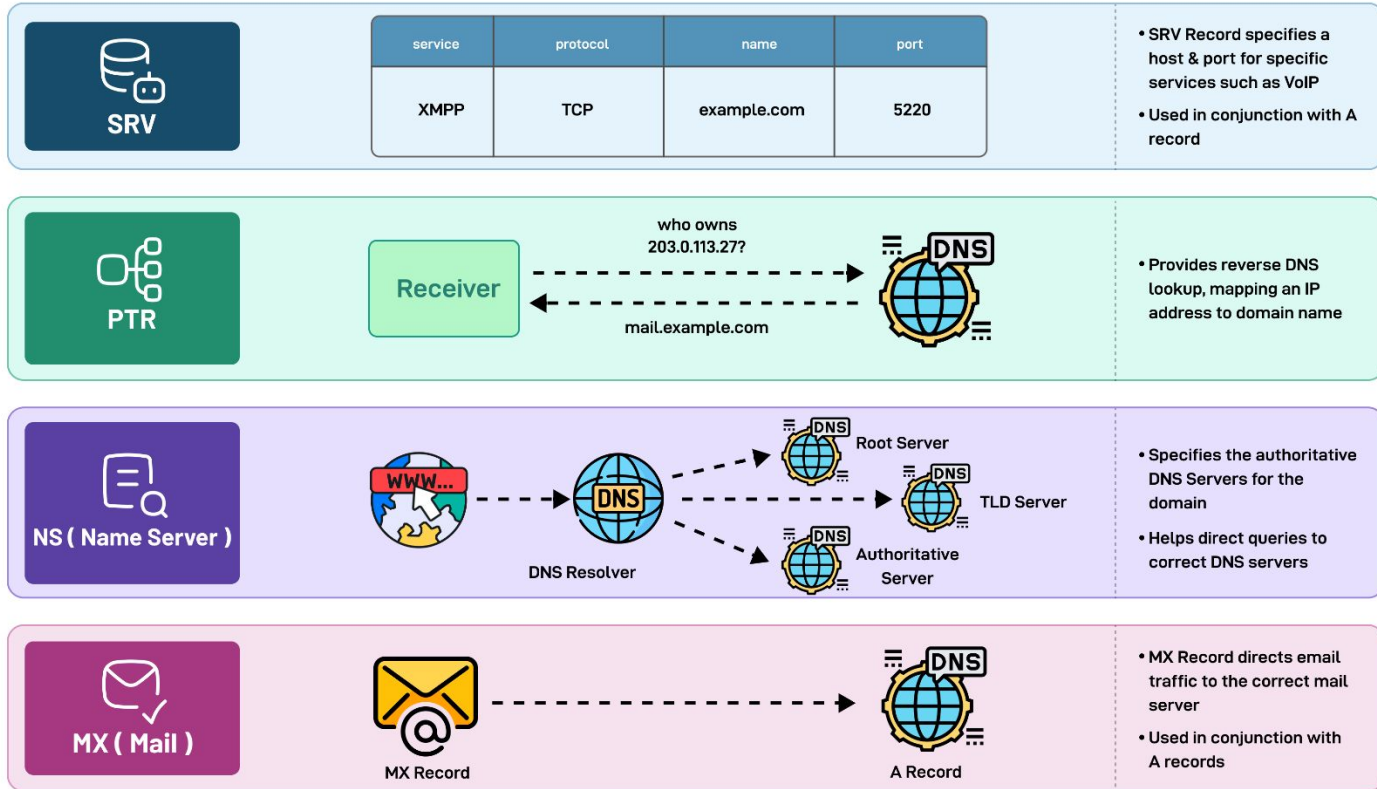
► Query and response messages: Header format

Identification	Flags
Number of question records	Number of answer records (all 0s in query message)
Number of authoritative records (all 0s in query message)	Number of additional records (all 0s in query message)

► DNS: Types of Records



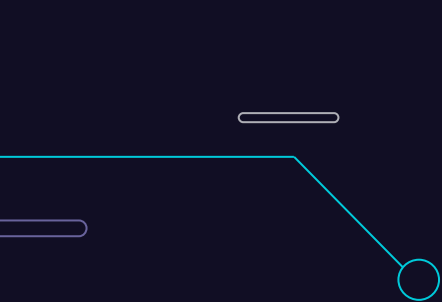
► DNS: Types of Records



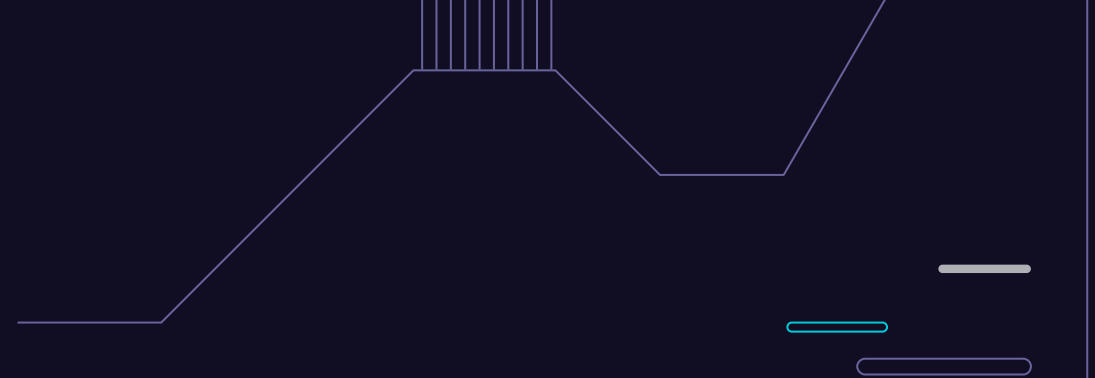
► DNS: Registrars

❖ How are new domains added to DNS?

- This is done **through a registrar**, a commercial entity accredited by **ICANN**.
- A registrar first **verifies** that the requested domain name is **unique** and then enters it into the **DNS database**.
- A fee is charged.



File Transfer and Email Protocols: FTP, SFTP, SMTP, IMAP, POP₃



6.4

► File Transfer Protocol (FTP):

- ❖ FTP is a **standard** internet protocol for **transmitting files** between computers on the Internet over **TCP/IP connections**.
- ❖ FTP is a **client-server protocol** that relies on two communications channels between client and server:
 - a **control channel** for controlling the conversation and
 - a **data channel** for transmitting file content.
- ❖ Clients **initiate** conversations with servers by requesting to download a file.
- ❖ Using FTP, a client can **upload, download, delete, rename, move** and **copy** files on a server.
- ❖ A user needs to log on to the FTP server or sometimes contents are available without login.

► File Transfer Protocol (FTP):

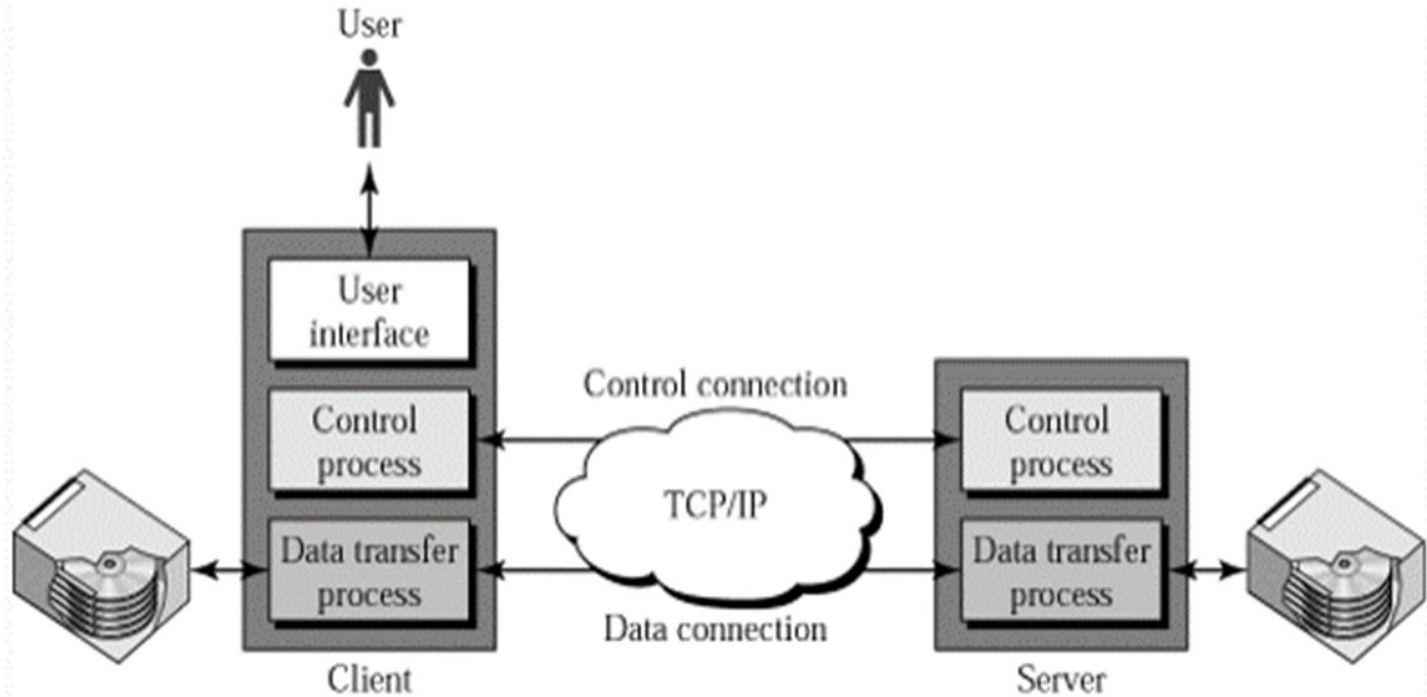


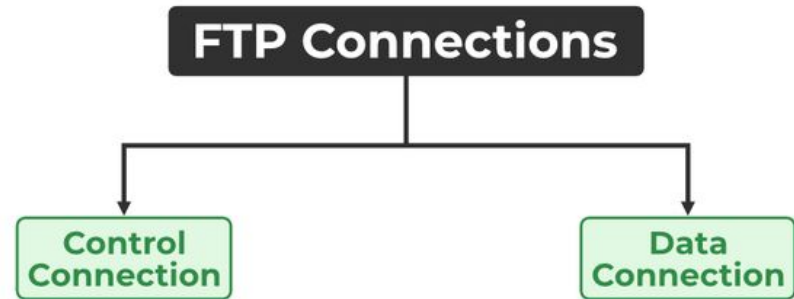
Fig:FTP

► File Transfer Protocol (FTP):

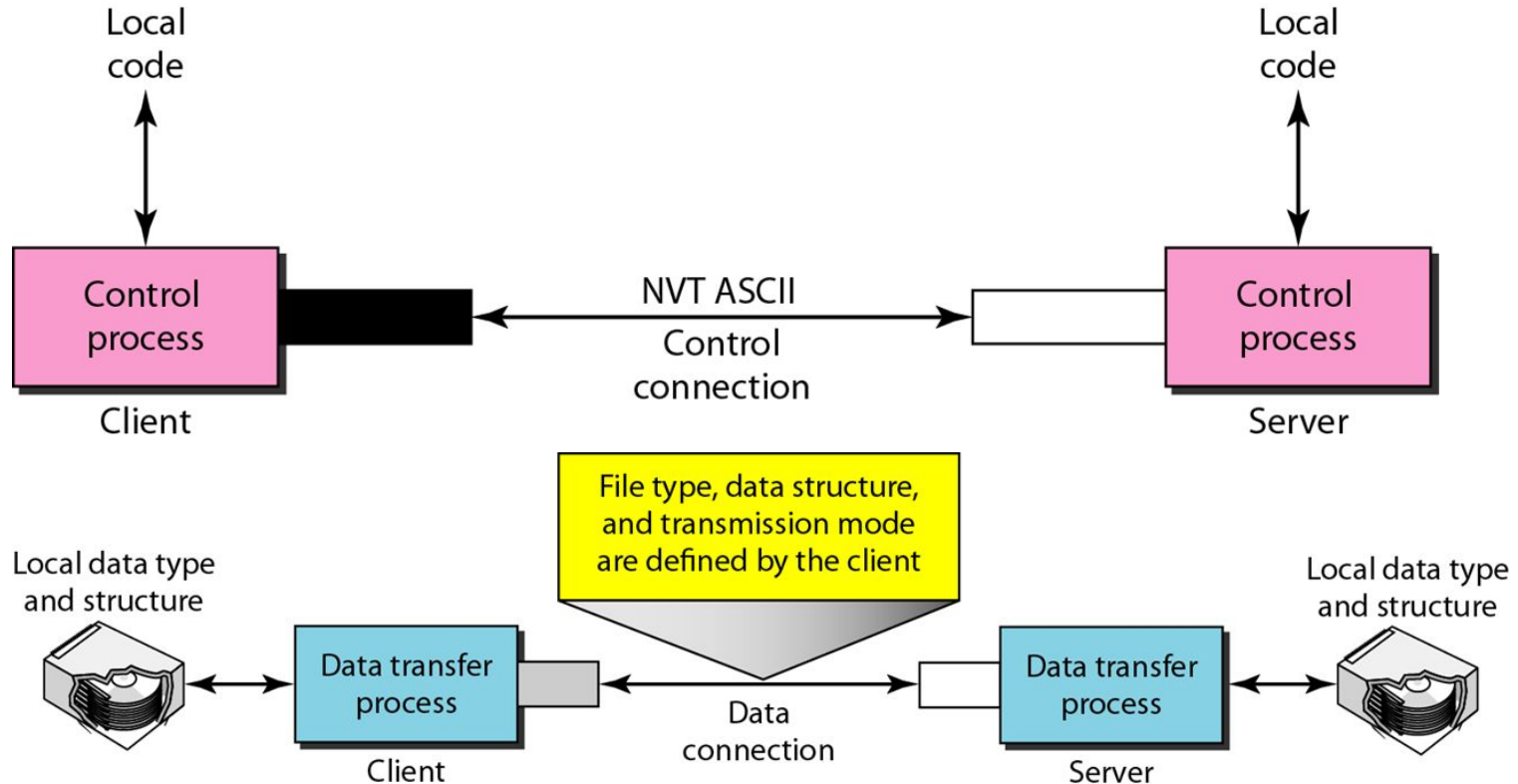
- ❖ FTP is the **standard mechanism** for one of the most common tasks on the Internet, **copying** a file from one computer to another.
- ❖ While transferring files, FTP **resolves** different name conventions, different representation of text and data or directory structures.
- ❖ It uses the Services of TCP.
- ❖ It needs Two TCP Connections.
 - The Well Known **Port 21** is used for **Control** Connection.
 - The Well Known **Port 20** is used for **Data** Connection.

► File Transfer Protocol (FTP):

- ❖ FTP **differs** from other client-server applications in that it establishes **two connections** between hosts.
- ❖ One connection is used for data transfer, the other for control information (commands and responses).
- ❖ Control connection transfers **only a line of** command or response at a time.
- ❖ Data connection follows **complex rules** due to the variety of data types transferred.



► FTP : Using the control and data connection



► FTP:

- ❖ **File Types:** FTP can transfer one of the following file types.

- ASCII file
- EBCDIC file (Extended Binary Coded Decimal Interchange Code)
- Image File

- ❖ **Data Structures:**

- File Structure (default)
- Record structure
- Page structure

- ❖ **Transmission Mode:**

- Stream Mode:
 - Continuous Stream of Bytes.
- Block Mode:
 - Blocks
- Compressed Mode:
 - Big files

▶ FTP:

◆ File Transfer:

➤ Retrieving File:

- Copying a file from **server to client.**
- [DOWNLOAD]

➤ Storing File:

- Copying a file from **client to Server.**
- [UPLOAD]

➤ Retrieving List:

- List of files/drive.
- From server to client.

◆ FTP Commands:

➤ Transfer Files:

- GET [DOWNLOAD]
- PUT [UPLOAD]

➤ Connect to Remote Host

- Open -open a connection
- User -authenticate username
- Pass- authenticate password

➤ Terminate Session

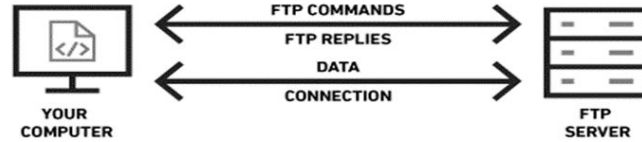
- **QUIT:** Disconnect from Remote Host and Terminate FTP
- **CLOSE:** Disconnect. NO Termination.

► SSH File Transfer Protocol(SFTP):

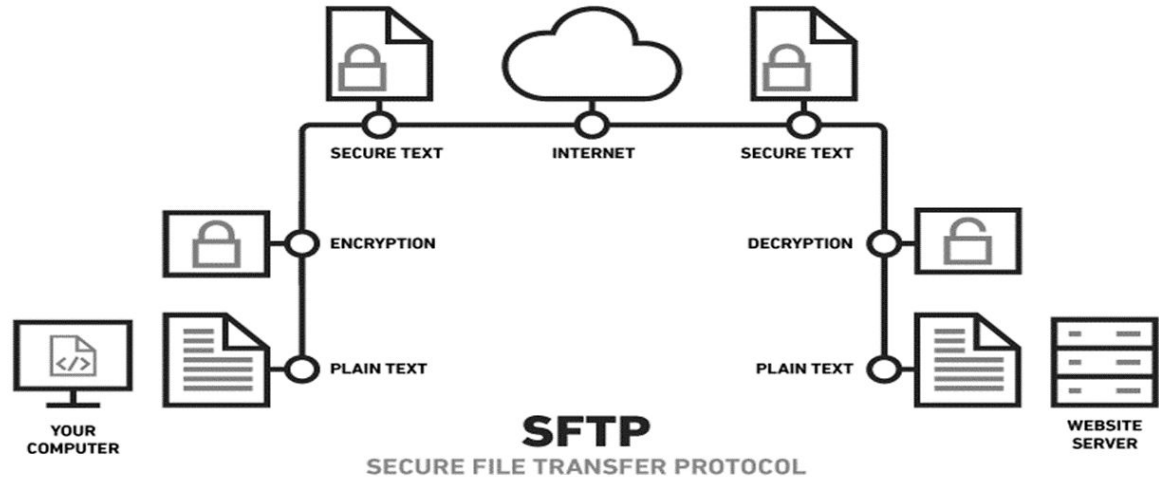
- ❖ It is a **secure file transfer** protocol.
- ❖ It runs over the **SSH protocol**.
- ❖ It supports the full security and authentication functionality of SSH.
- ❖ SFTP has pretty much **replaced** FTP.
- ❖ SFTP provides more security and reliability and easier to configure.
- ❖ Protects the integrity of data using **encryption** and **cryptographic** techniques and **authenticates** both server and user.

► SSH File Transfer Protocol (SFTP):

FTP FILE TRANSFER PROTOCOL



FTP VS SFTP



► SSH Protocol:

- ❖ The SSH protocol (also referred to as **Secure Shell**) is a method for secure **remote login** from one computer to another.
- ❖ It provides several alternative options for strong authentication, and it protects the communications **security and integrity** with strong encryption.
- ❖ It is a secure alternative to the **non-protected login protocols** (such as telnet, rlogin) and **insecure file transfer** methods (such as FTP).

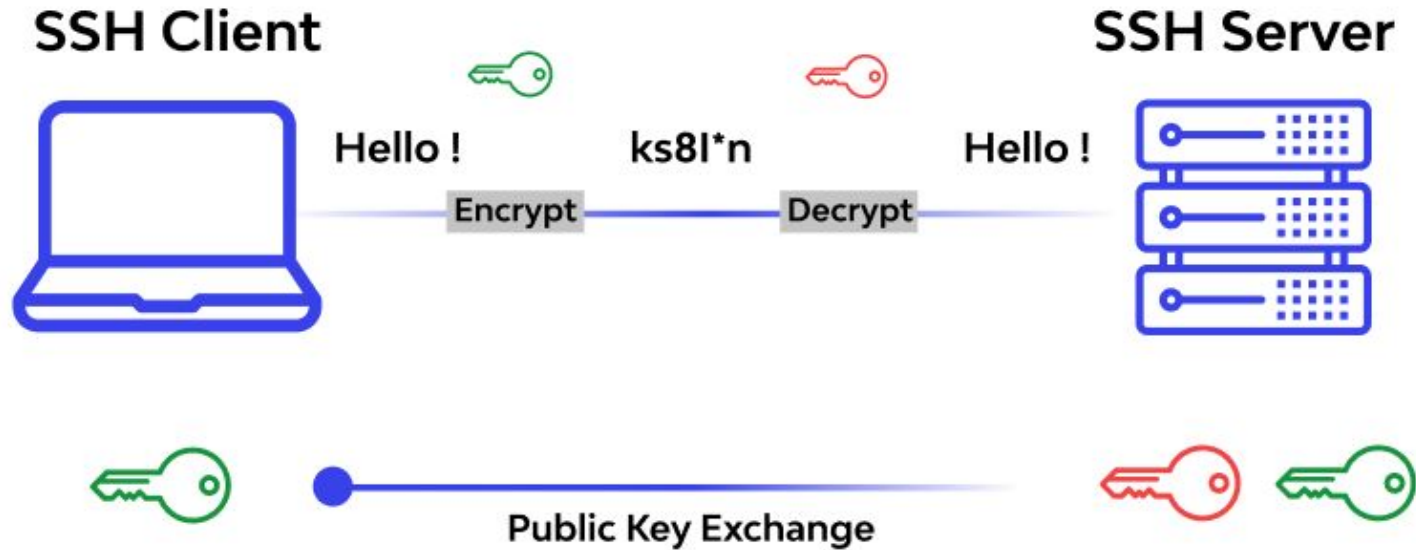
► TYPICAL USES OF THE SSH PROTOCOL:

- ❖ The protocol is used in corporate networks for:
 - Providing **secure access** for users and automated processes
 - Interactive and automated file transfers
 - Issuing **remote** commands
 - Managing network infrastructure and other mission-critical system components.

► HOW DOES THE SSH PROTOCOL WORK:

- ❖ The protocol works in the **client-server model**, which means that the connection is established by the **SSH client** connecting to the **SSH server**.
- ❖ The SSH client drives the connection setup process and uses **public key** cryptography to verify the **identity** of the **SSH server**.
- ❖ After the setup phase the SSH protocol uses strong **asymmetric encryption** and **hashing** algorithms to ensure the privacy and integrity of the data that is exchanged between the client and server.

► HOW DOES THE SSH PROTOCOL WORK:



► Electronic Mail:

- ❖ One of the most popular Internet services is **electronic mail** (e-mail).
- ❖ The designers of the Internet probably never imagined the popularity of this application program.
- ❖ Its **architecture consists of** several components
 - User Agent
 - **Message Transfer Agent:** SMTP
 - **Message Access Agent:** POP and IMAP
- ❖ Email is emerging as one of the most valuable services on the internet today.
- ❖ Most of the internet systems use **SMTP** as a method to transfer mail from one user to another.

► Electronic Mail:

- ❖ SMTP is a **push protocol** and is used to **send the mail** whereas POP (post office protocol) or IMAP (internet message access protocol) are used to **retrieve those mails** at the receiver's side.
- ❖ **User Agent :**
 - In computing, a **user agent** is software (a software agent) that is acting on behalf of a user, such as a **web browser** that "retrieves, renders and facilitates end user interaction with Web content".
 - An **email reader** is a mail user agent.
 - In many cases, a user agent **acts as a client** in a network protocol used in communications within a client-server distributed computing system.
 - In particular, the Hypertext Transfer Protocol (HTTP) **identifies** the client software originating the request, using a **user-agent header**, even when the client is not operated by a user.

► Electronic Mail:

❖ Mail Transfer Agent:

- Within the Internet email system, a message transfer agent or mail transfer agent (MTA) or mail relay is software that **transfers** electronic mail messages **from one computer** to **another** using SMTP.
- The terms mail server, mail exchanger, and MX host are also used in some contexts.
- Messages exchanged across networks are passed **between mail servers**, including any attached data files (such as images, multimedia or documents)
- It is a software application used within an **Internet message handling system (MHS)**. It is responsible for **transferring and routing** an electronic mail message from the sender's computer to the recipient's computer.
- The basic platform for an MTA is **an exchange system** with client/server architecture.
- A message transfer agent receives incoming emails and forwards the messages to individual clients/users. The main function of the MTA is **forwarding** the incoming message to the proper end-user or destination.
- **Microsoft Exchange** and **UNIX** send mail are the best examples of MTAs.

► Mailing Protocols: SMTP

- ❖ SMTP is an **application layer** protocol.
- ❖ The client who wants to send the mail **opens** a TCP connection to the **SMTP server** and then sends the mail across the connection.
- ❖ The SMTP server is **always on listening mode**. As soon as it listens for a TCP connection from any client, the SMTP process **initiates a connection** on that **port (25)**.
- ❖ After **successfully establishing** the TCP connection the client process **sends** the mail instantly.
- ❖ The SMTP model is of **two type** :
 - End-to- end method
 - Store-and- forward method

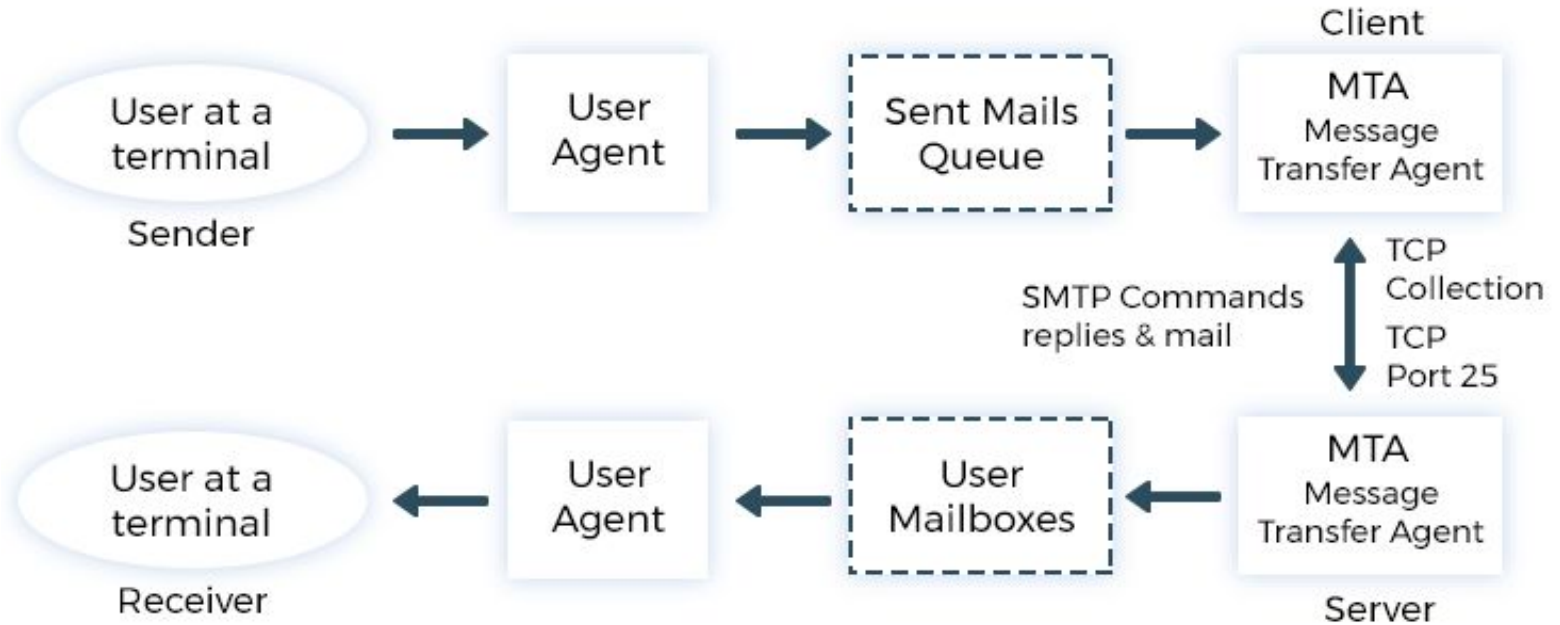
► Mailing Protocols:SMTP

- ❖ The **end to end model** is used to communicate **between different organizations** whereas the **store and forward** method are used **within** an organization.
- ❖ A **SMTP client** who wants to send the mail will contact the **destination's host SMTP** directly in order to send the mail to the destination.
- ❖ The **SMTP server** will keep the mail to itself until it is **successfully copied** to the receiver's SMTP.
- ❖ The **client SMTP** is the one which **initiates** the session let us call it as the client- SMTP and the **server SMTP** is the one which **responds** to the session request and let us call it as receiver-SMTP.
- ❖ The client- SMTP will start the session and the receiver-SMTP will respond to the request.

► Model of SMTP System:

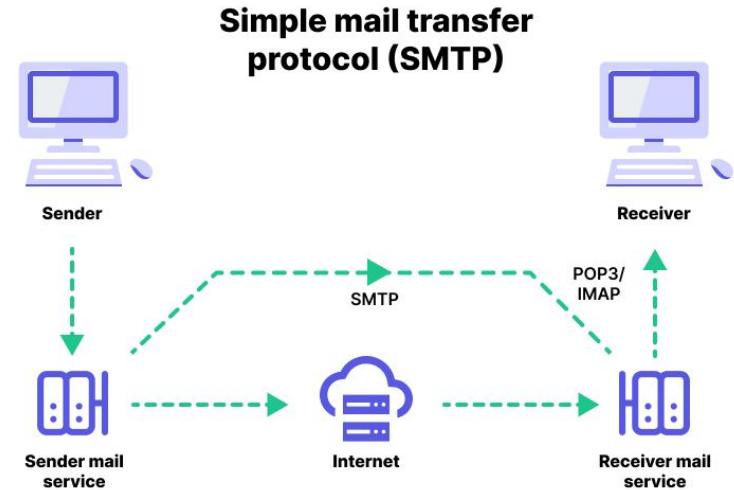
- ❖ In the SMTP model **user deals** with the **user agent (UA)** for example Microsoft Outlook, Netscape, Mozilla, etc.
- ❖ In order to **exchange the mail** using TCP, MTA is used.
- ❖ The users sending the mail **do not** have to **deal** with the **MTA** it is the responsibility of the **system admin** to set up the local MTA.
- ❖ The MTA maintains a **small queue** of **mails** so that it can schedule **repeat delivery** of mail in case the receiver is **not available**.
- ❖ The MTA delivers the mail to the mailboxes and the information can **later be downloaded** by the user agents.

► Model of SMTP System:



► Communication between sender and the receiver

- ❖ The senders, **user agent** prepare the message and **send it to the MTA**.
- ❖ The **MTA** functioning is to **transfer the mail** across the network to the **receivers MTA**.
- ❖ To send mail, a system must have the **client MTA**, and to receive mail, a system must have a **server MTA**.



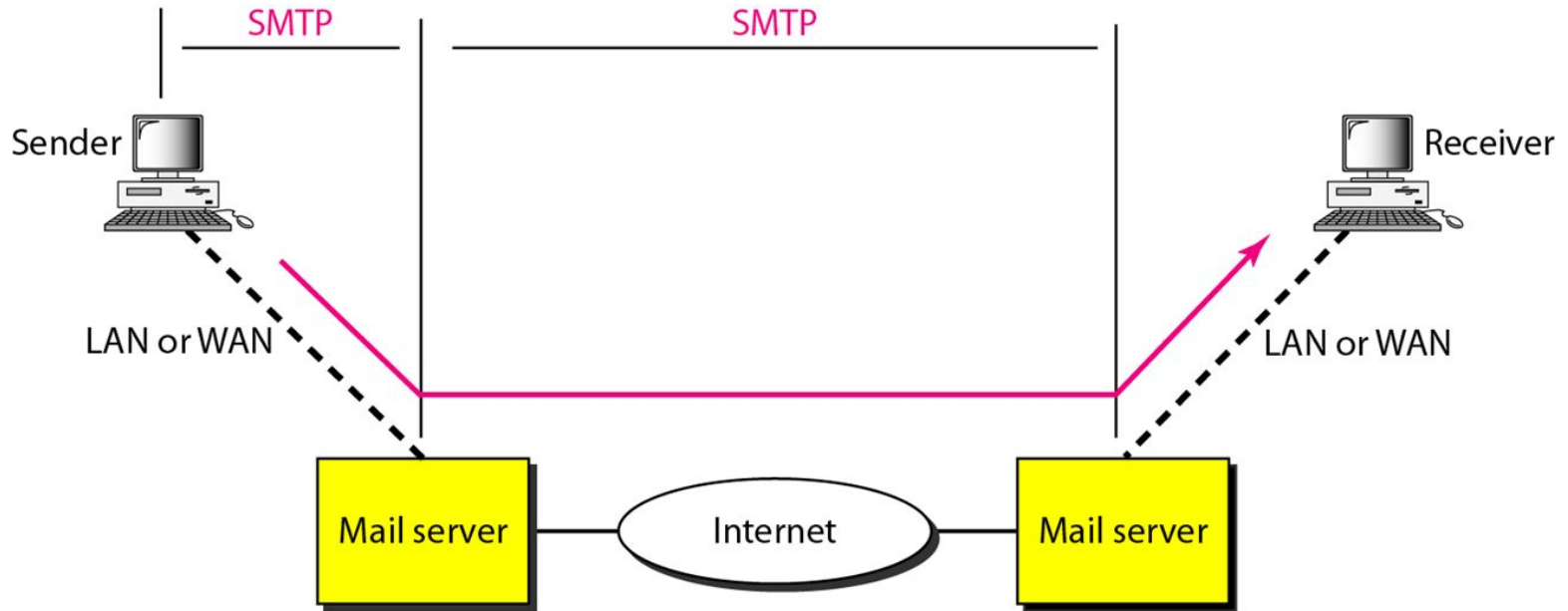
► SENDING EMAIL:

- ❖ Mail is sent by a **series of** request and response messages between the client and a server.
- ❖ The **message** which is sent across consists of a **header** and **the body**.
- ❖ Everything which is **after the header** is considered as the **body** of the message which is a **sequence** of ASCII characters.
- ❖ The message body contains the **actual information** read by the receipt.

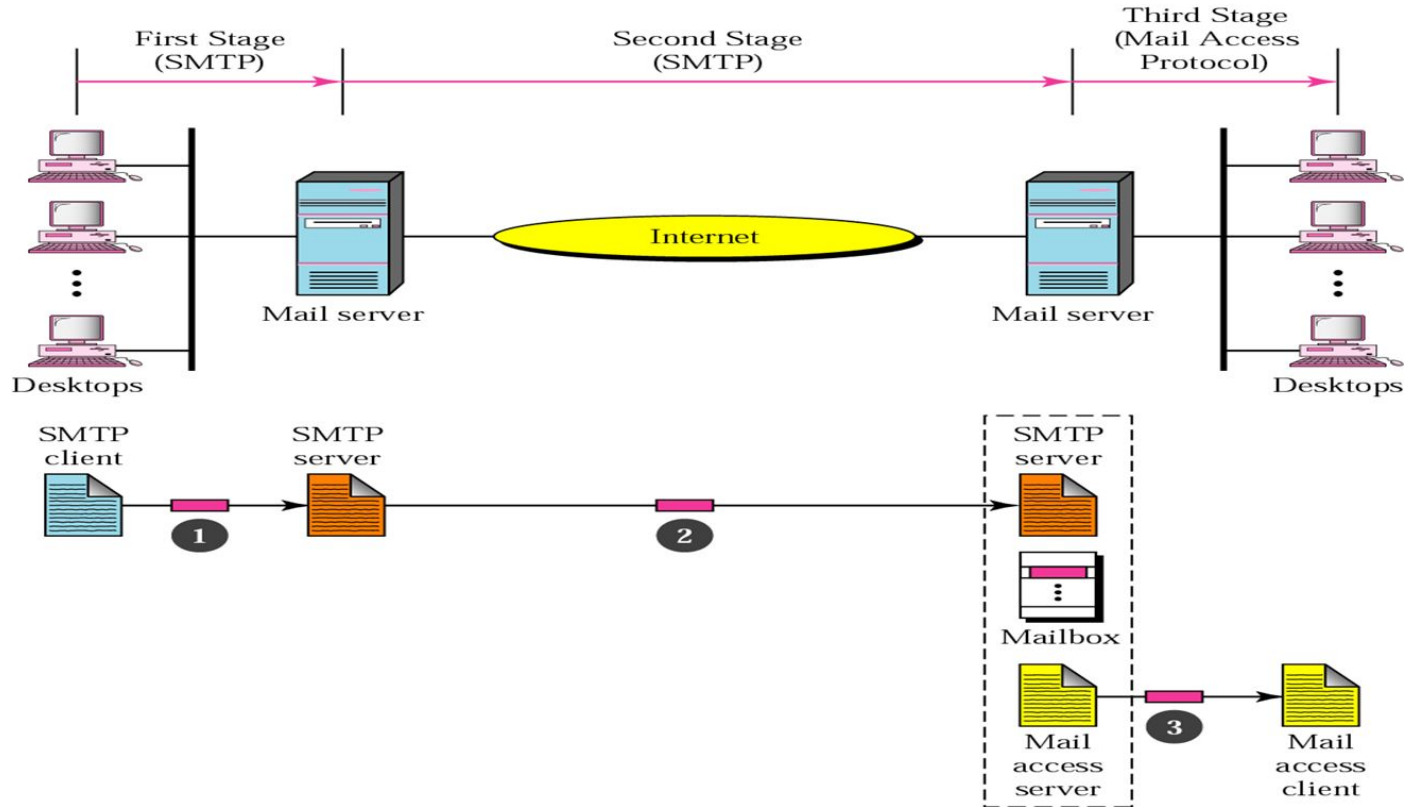
► RECEIVING EMAIL:

- ❖ The **user agent** at the **server side** checks the **mailboxes** at a particular time of **intervals**.
- ❖ If any information is received **it informs the user** about the mail.
- ❖ When the user tries to read the mail it **displays a list of mails** with a short description of each mail in the **mailbox**.
- ❖ By selecting any of the mail user can view its **contents** on the **terminal**.

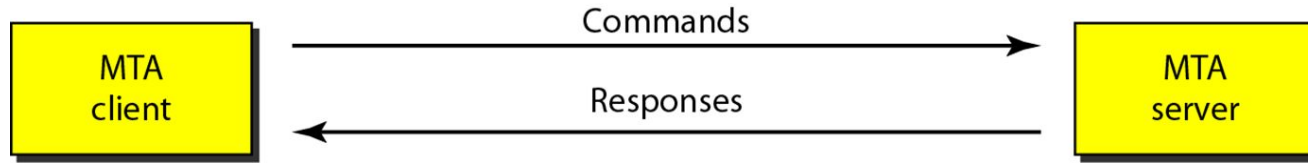
► SMTP :Range



► SMTP : Example



► SMTP : Commands and responses



► SMTP : Commands

<i>Keyword</i>	<i>Argument(s)</i>
HELO	Sender's host name
MAIL FROM	Sender of the message
RCPT TO	Intended recipient of the message
DATA	Body of the mail
QUIT	
RSET	
VERFY	Name of recipient to be verified
NOOP	
TURN	
EXPN	Mailing list to be expanded
HELP	Command name
SEND FROM	Intended recipient of the message
SMOL FROM	Intended recipient of the message
SMAL FROM	Intended recipient of the message

► SMTP : Responses

Code	Description
Permanent Negative Completion Reply	
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

Code	Description
Positive Completion Reply	
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
Positive Intermediate Reply	
354	Start mail input
Transient Negative Completion Reply	
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted: insufficient storage

► Electronic Mail: First Scenario

UA: user agent

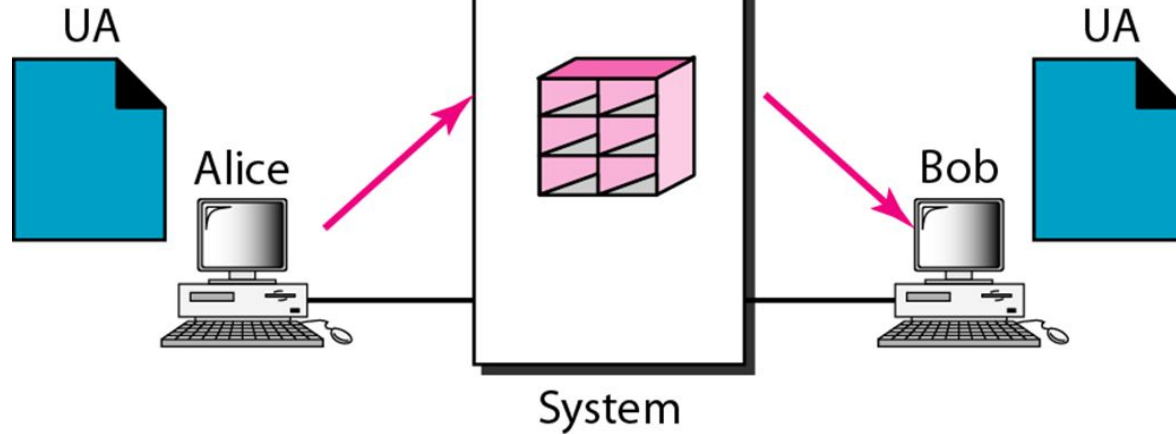
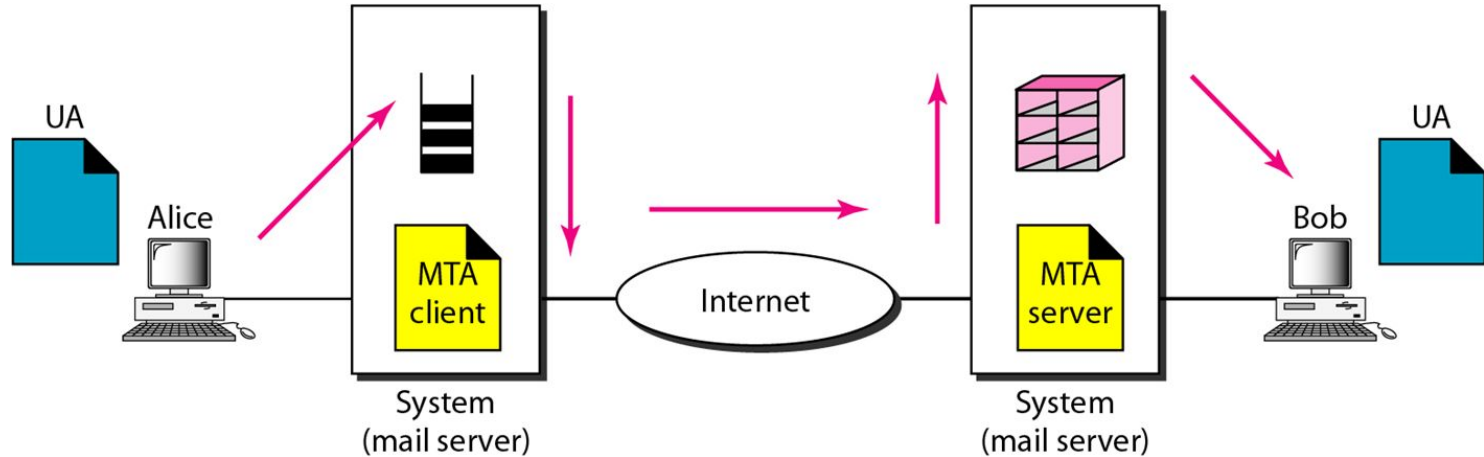


Figure: When the sender and the receiver of an e-mail are on the same system, we need only two user agents.

► Electronic Mail : Second Scenario

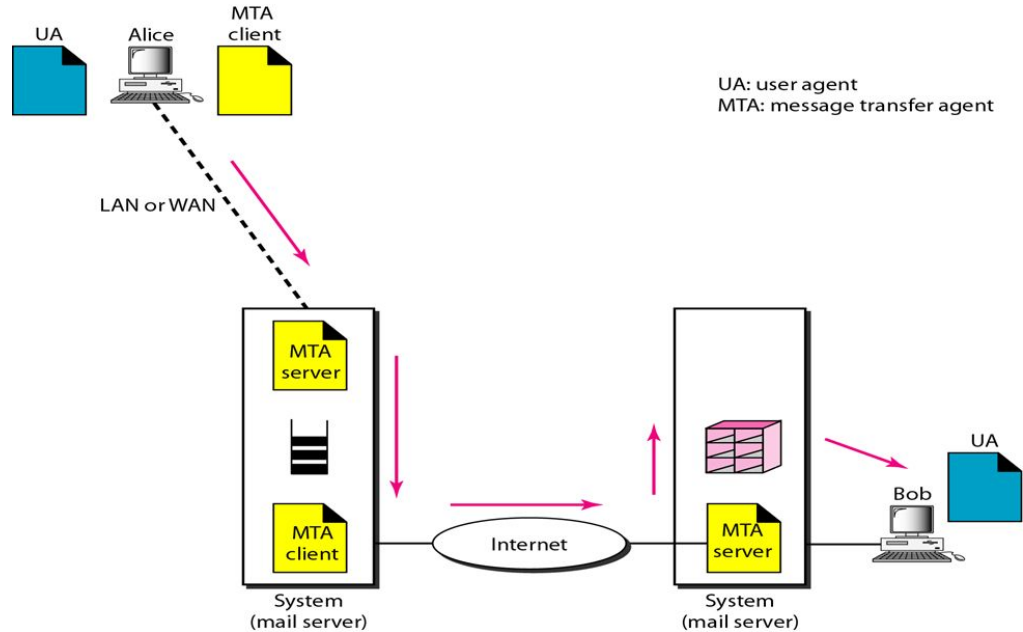
UA: user agent

MTA: message transfer agent



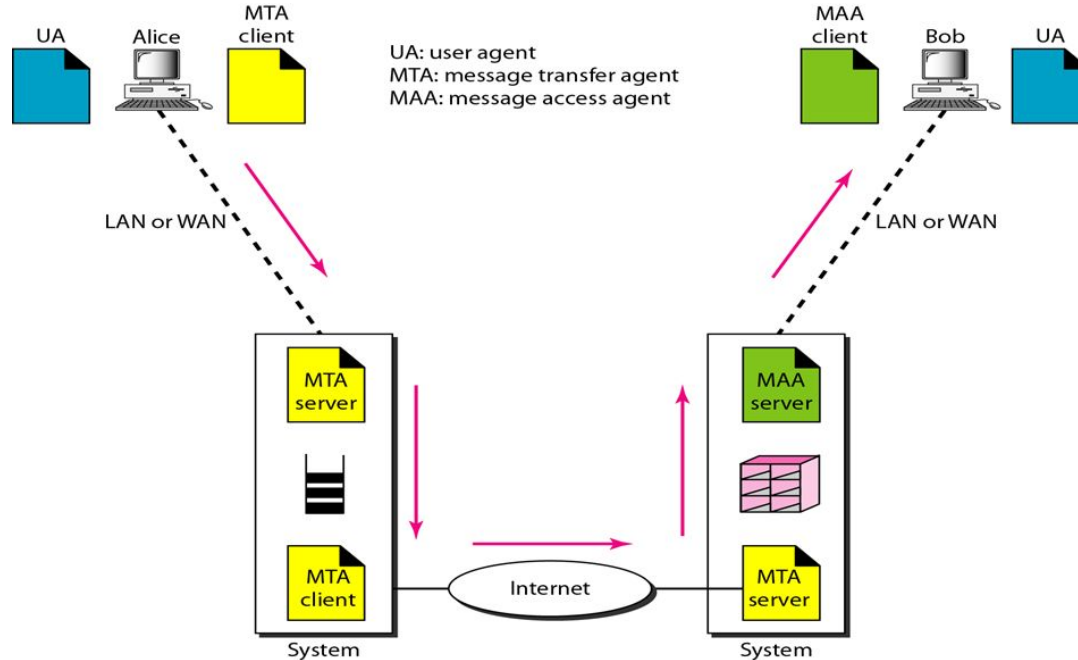
When the sender and the receiver of an e-mail are on different systems, we need two UAs and a pair of MTAs (client and server).

► Electronic Mail : Third Scenario



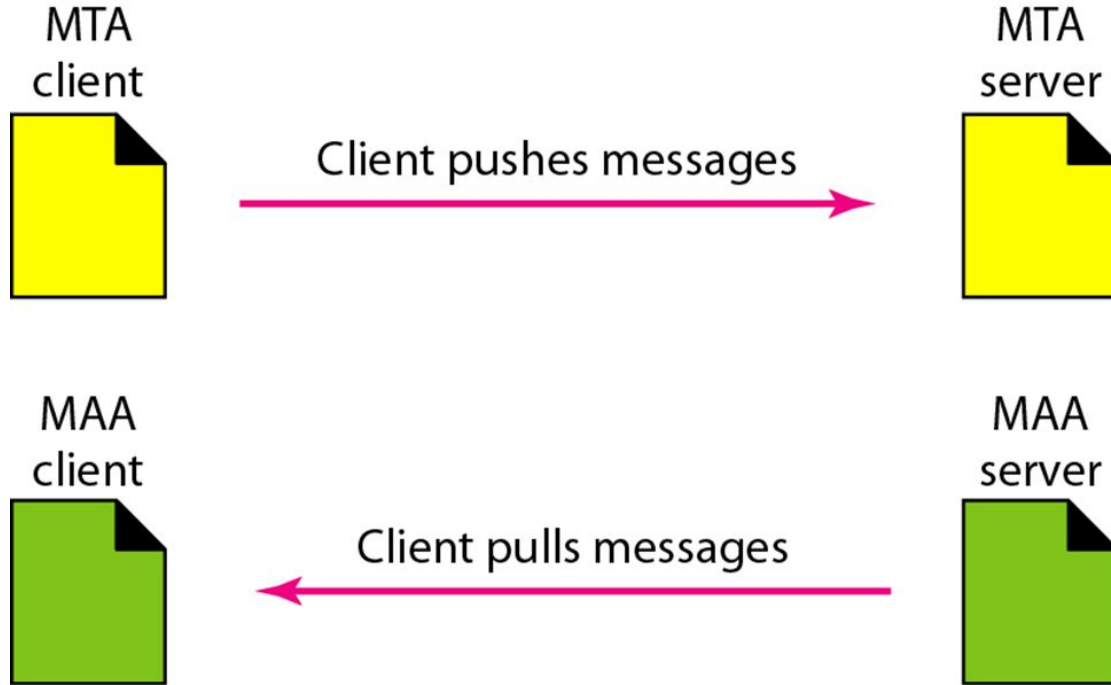
When the sender is connected to the mail server via a LAN or a WAN, we need two UAs and two pairs of MTAs (client and server).

► Electronic Mail : Fourth Scenario

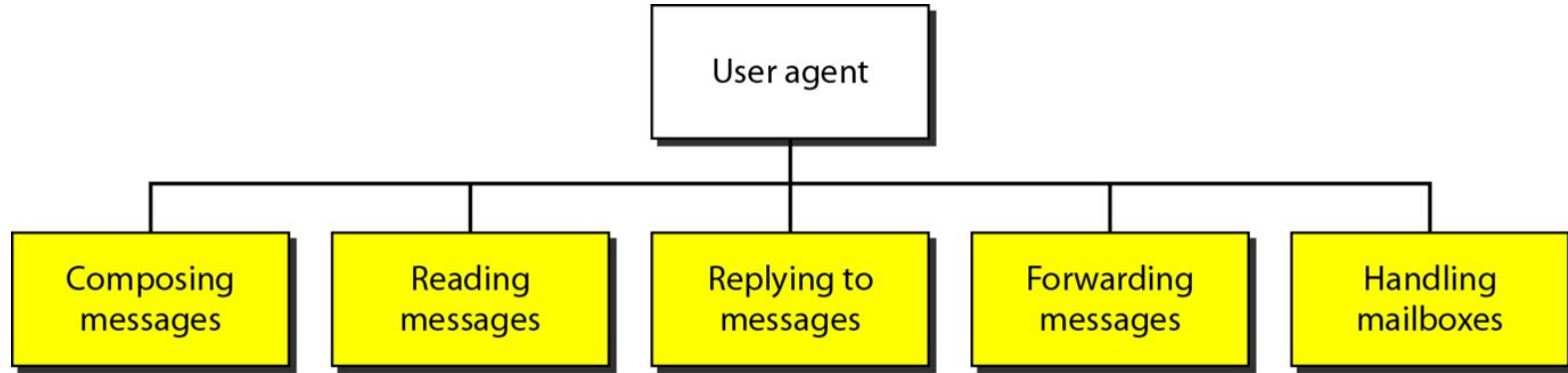


When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs and a pair of MAAs. This is the most common situation today.

► Electronic Mail: Push versus pull

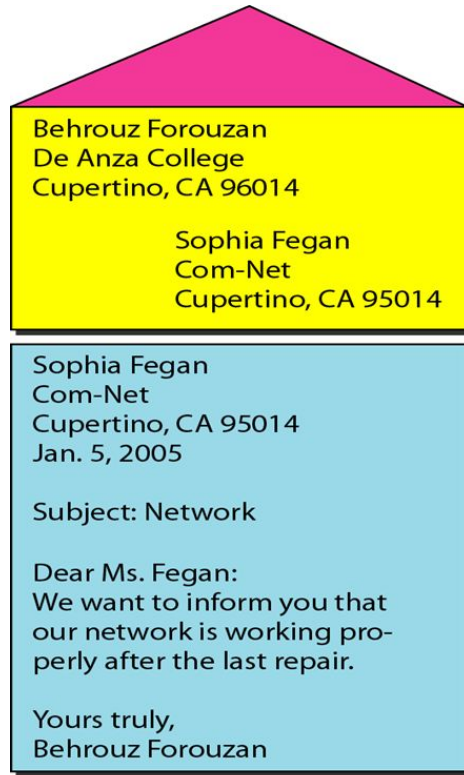


► Electronic Mail : Services of user agent

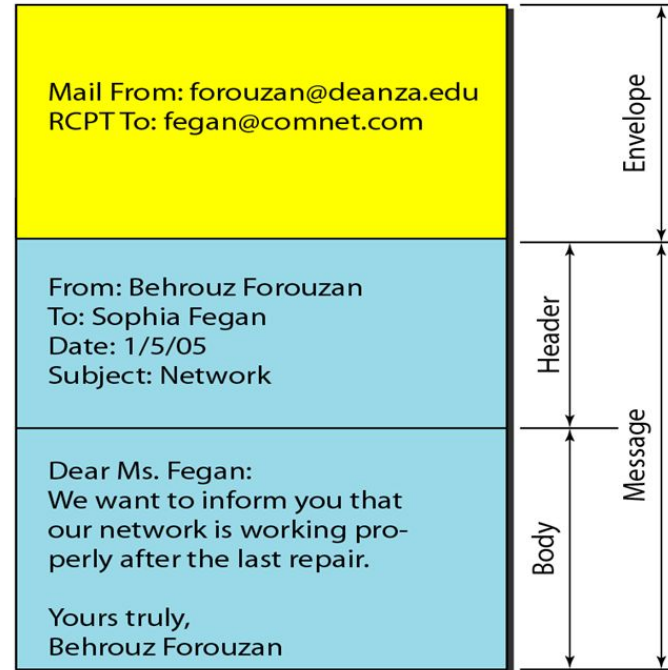


- ❖ Some examples of command-driven user agents are mail, pine, and elm.
- ❖ Some examples of GUI-based user agents are Eudora, Outlook, and Netscape.

► Electronic Mail: Format

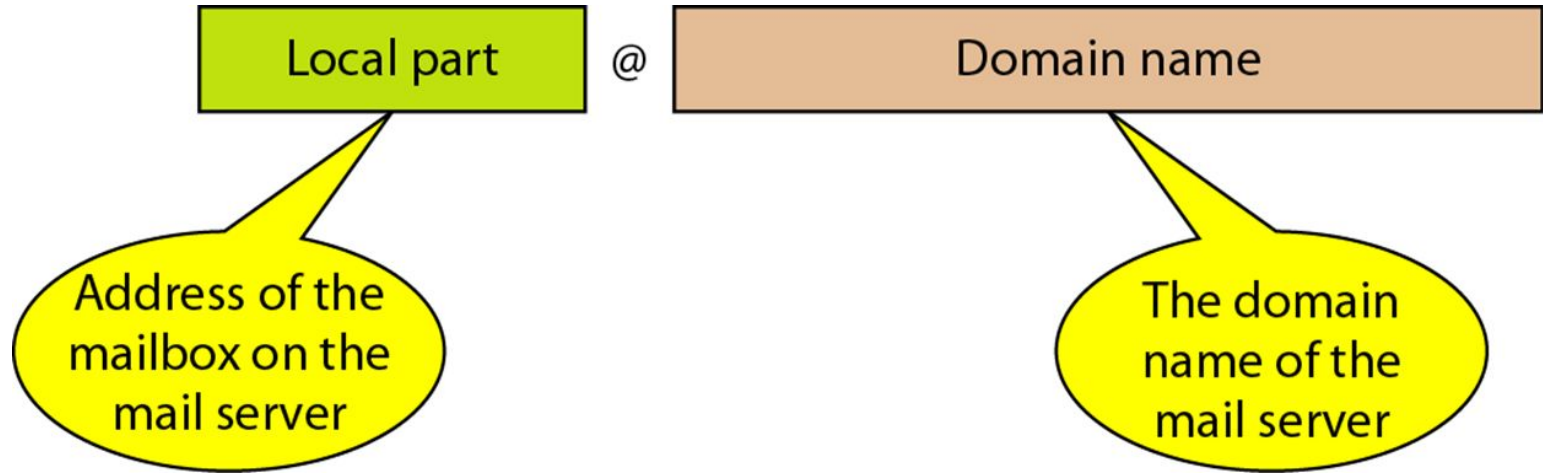


a. Postal mail



b. Electronic mail

► Electronic Mail : Address



► Post Office Protocol(POP):

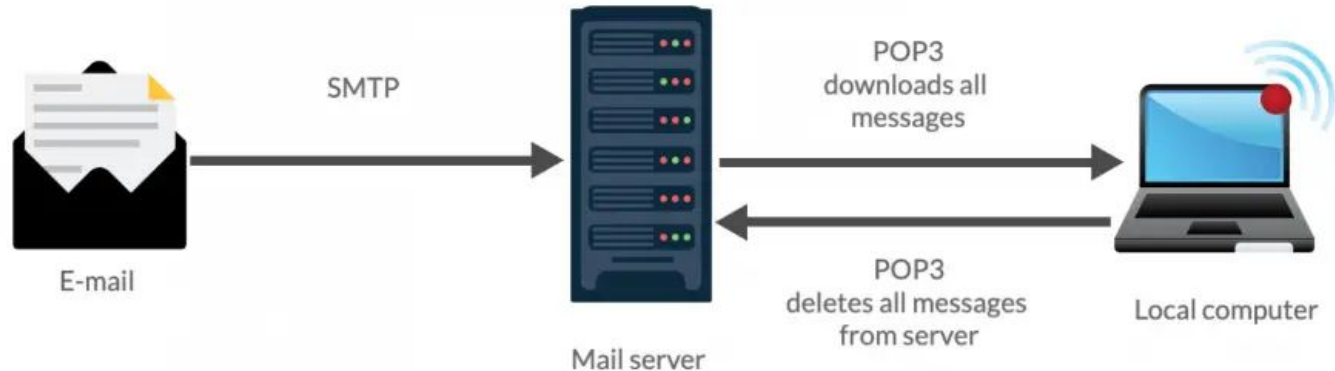
- ❖ Standard protocol **used to receive emails** from a remote server to a **local mail client**.
- ❖ Allows to **download** email messages on **user's local computer** and read them **offline**.
- ❖ When using POP, messages are downloaded locally and **removed from** the mail server.
- ❖ If using mail account from **multiple** locations, its **not the best option**.
- ❖ Since, messages are stored on local computer, it **reduces** the space on email account.
 - By default, POP works on **two ports**:
 - **Port 110:**
 - ◆ Default non-encrypted port
 - **Port 995:**
 - ◆ secure POP3 port
- ❖ Simple protocol with limited functionality.
- ❖ **Client POP** software is installed on recipient computer and **server POP** software is installed on mail server.

► Working of POP3:

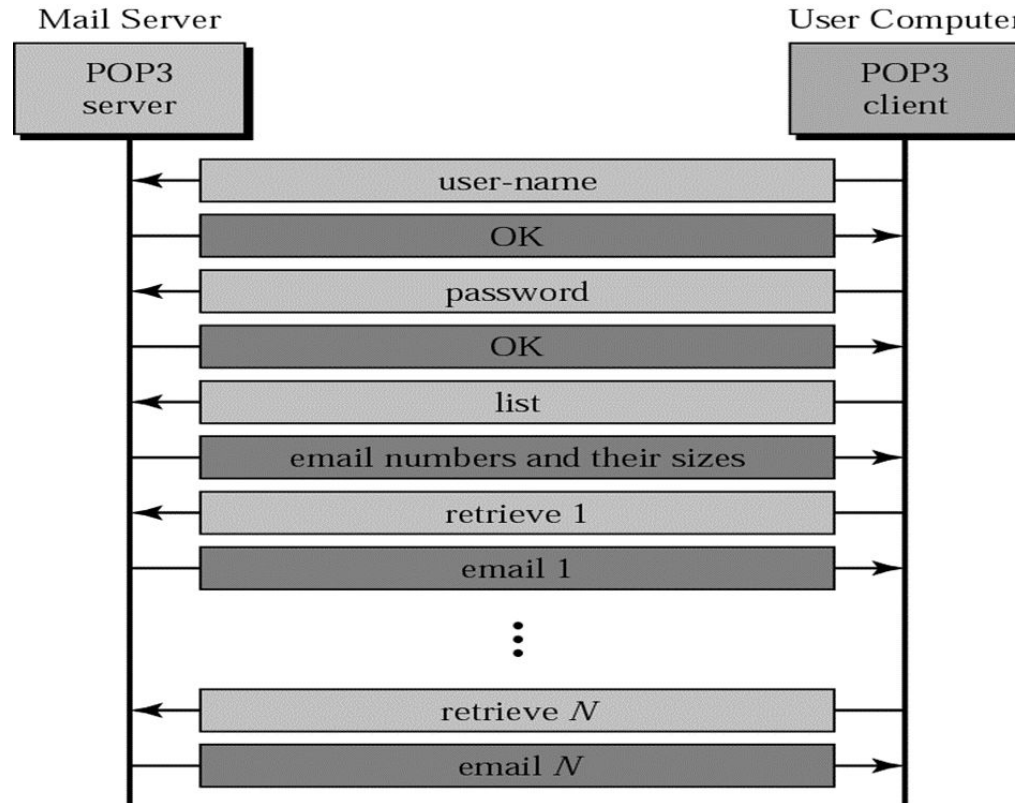
- ❖ **Mail access** starts with the client when the user needs to **download** the received email from the **mailbox** on the **mail server**.
- ❖ The client(user agent) sends the **username and password** to access the **mailbox**.
- ❖ The user can then **list** and **retrieve** the mail messages one by one.
- ❖ Application Layer Protocol used by Email Clients to retrieve Email From Remote Server.
- ❖ POP Version 3 is referred as POP3.
- ❖ A POP3 Servers Listens on Well Known **Port 110**.
- ❖ Encryption Communication for POP3 Using SSL.
- ❖ It uses Well Known **TCP Port 995** (Eg. Google Gmail).

► POP3 : Post Office Protocol Version 3

How POP3 works?



► POP3 : Post Office Protocol Version 3 (Steps)



► Internet Message Access Protocol(IMAP):

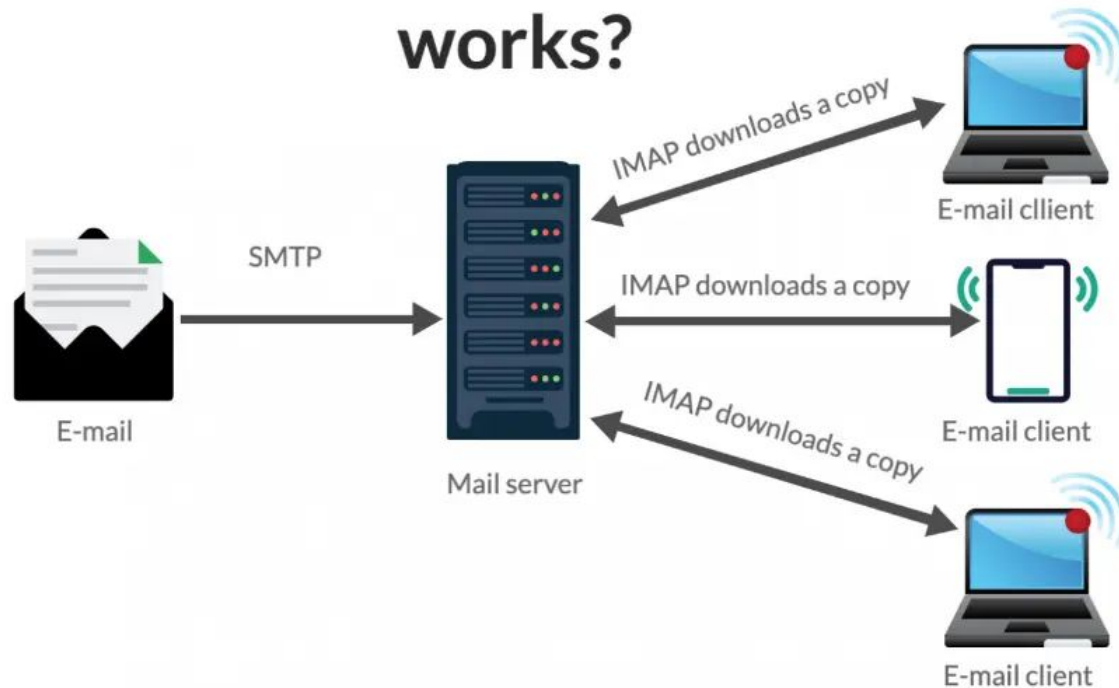
- ❖ Internet Message Access Protocol (IMAP) is a standard protocol for **accessing email** on a **remote server** from a **local client**.
- ❖ IMAP is an application layer Internet Protocol using the underlying **transport layer protocols** to establish **host-to-host communication** services for applications.
- ❖ This allows the use of a remote mail server. The well-known port address for **IMAP is 143**.
- ❖ The IMAP architecture enables users to send and receive emails through a remote server, **without support** from a particular device.
- ❖ This type of email access is **ideal for travelers** receiving or answering emails from their home desktop or office computer.
- ❖ This term is also known as interactive mail access protocol, Internet mail access protocol.

► Internet Message Access Protocol(IMAP):

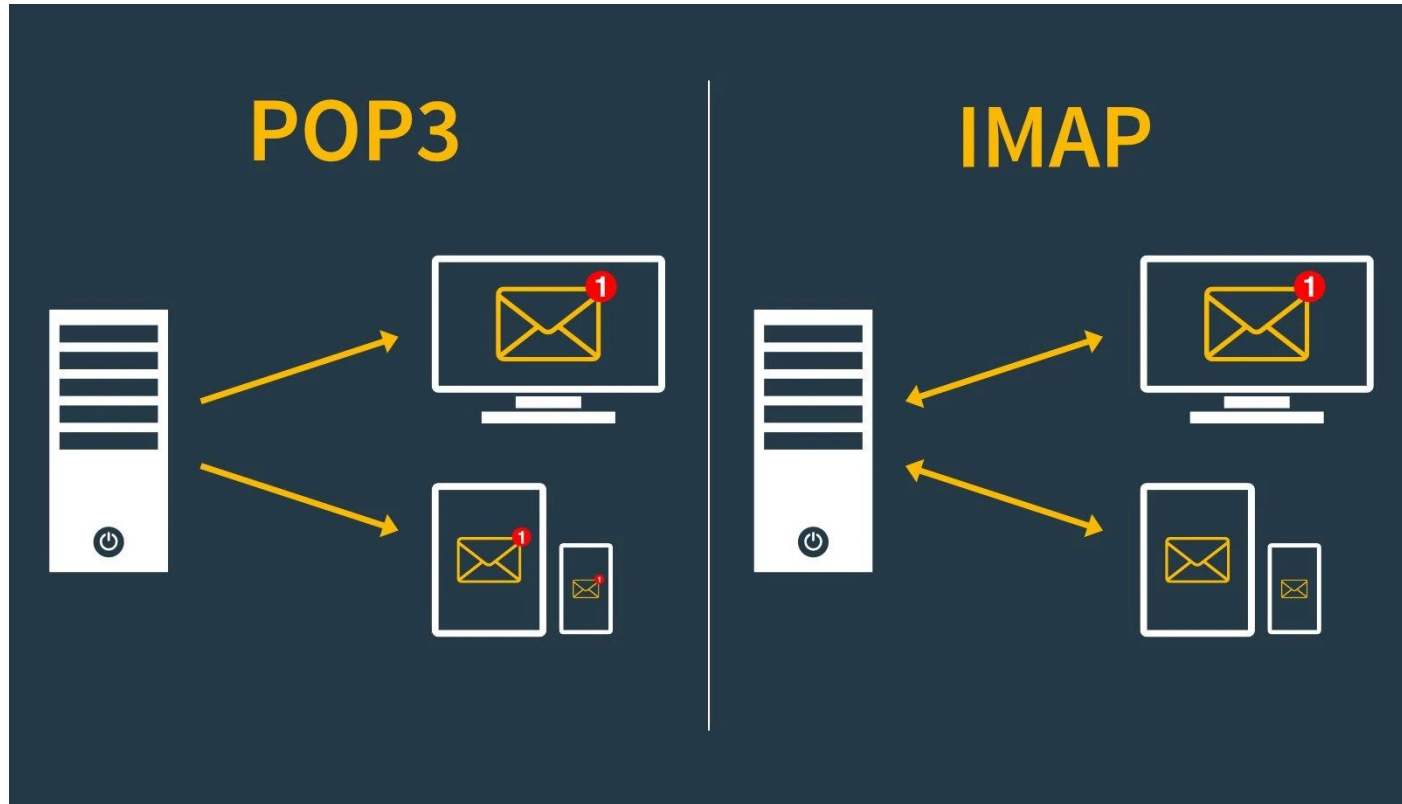
- ❖ IMAP is a **remote file server**, while POP **stores and forwards**.
- ❖ In other words, with IMAP, all emails **remain on the server** until the **client deletes** them.
- ❖ IMAP also permits multiple clients to **access and control** the **same** mailbox.
- ❖ When a user requests an email, it is routed through a central server.
- ❖ This keeps a storage document for the email files. Some of IMAP benefits include the ability to **delete messages**, search for **keywords** in the body of emails, create and manage **multiple mailboxes** or folders, and view the headings for **easy visual scans** of emails.
- ❖ IMAP is still used extensively, but is **less important** now that so much email is sent via web-based interfaces such as gMail, Hotmail, Yahoo Mail, etc.

► Internet Message Access Protocol(IMAP):

How IMAP works?

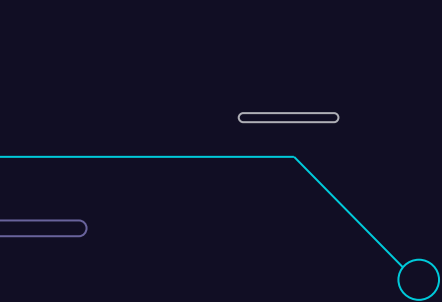


► IMAP vs POP3:

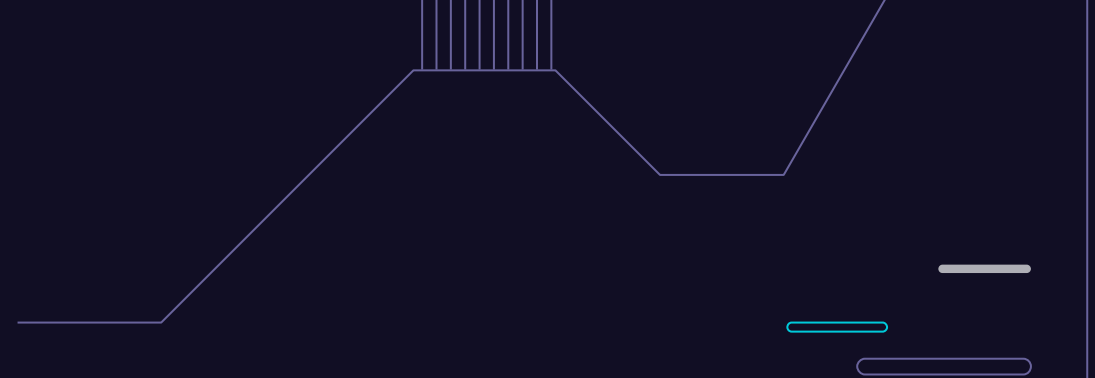


▶ IMAP VS POP3:

Feature	IMAP	POP3
Synchronization	Synchronizes changes across all devices	Changes made do not reflect on other devices
Storage	Emails stored on the server	Emails downloaded and often deleted from server
Access	Access to old mail and server-stored folders	Once downloaded, mail is typically removed from the server
Offline Availability	Works best with an active internet connection, some emails cached for offline viewing	Emails are downloaded to the device and can be viewed offline
Multiple Devices	Ideal for users accessing email on multiple devices	Better suited for single device access
Impact of Data Loss	Lower risk as data is stored on server	Higher risk as data is stored locally, and loss of device can mean loss of emails
Server Space	Can consume significant server space.	Frees up server space after email retrieval.



Overview of Application Server Concepts: Proxy, Web, Mail

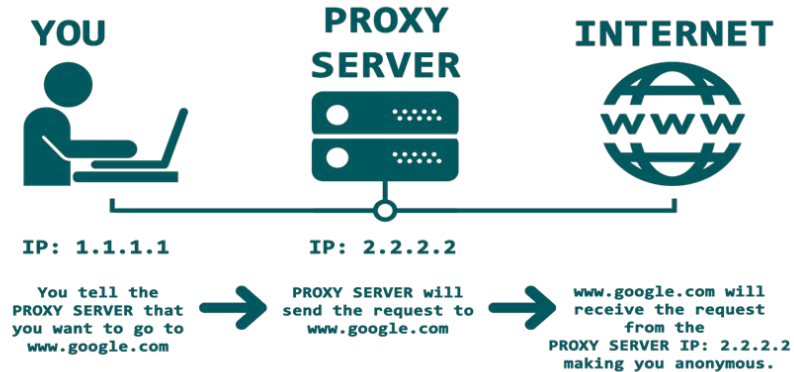
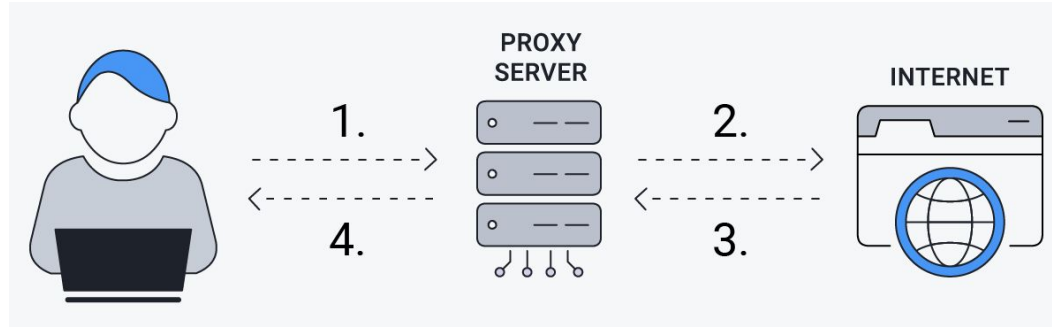


6.5

► Proxy Application Server:

- ❖ Acts as a **gateway** between user and the internet.
- ❖ Provide various levels of functionality, **security** and **privacy**.
- ❖ If using proxy server, internet traffic **flows through proxy server** on its way to the address requested.
- ❖ The request then **comes back** through that **same proxy server** and then the proxy server **forwards** the data received from the website to user.
- ❖ **Advantage**
 - Its cache can serve several users.
 - Improves the response time of website.

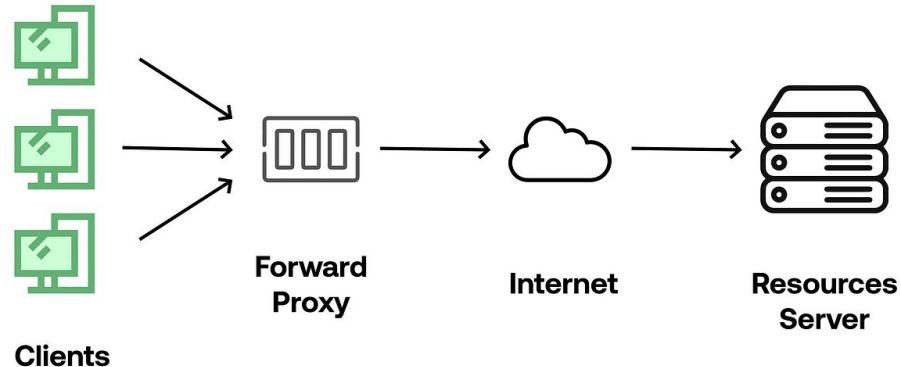
► Proxy Application Server:



► Types of Proxy Servers:

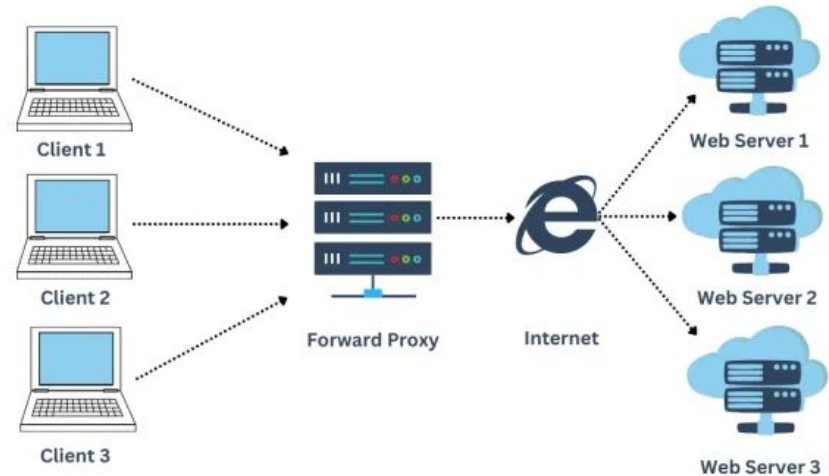
◆ Forward Proxy Servers

- A forward proxy, also known as a regular proxy or proxy server, is used to **receive and forward client connection** requests to a **destination web server** through a server.
- **used** by the **client** to bypass **firewall restrictions** in order to visit websites that are **blocked** by organization.
- **hides the real IP** of the client so that user can visit website that are blocked.



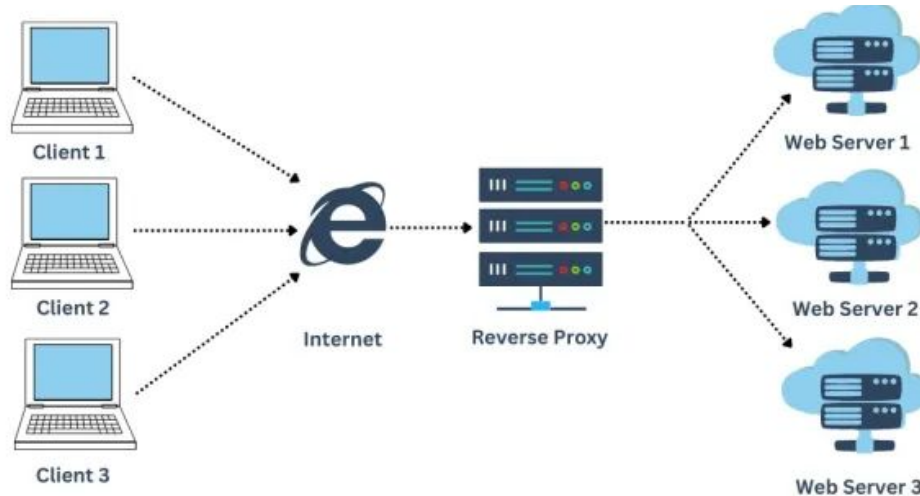
► Forward Proxy

- ❖ A forward proxy server **sits between** clients and the internet to establish a connection with the internet on **behalf of** the clients.
- ❖ If the client's connection **request is valid**, the forward proxy server will **forward** the request to the **destination** on the client's behalf.
- ❖ If the client's **connection request is invalid**, the forward proxy server will **redirect it** to the **client without** processing it.



► 2.Reverse Proxies

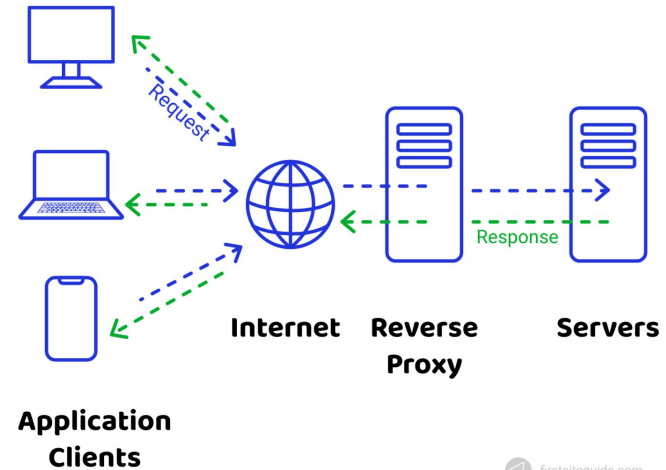
- ❖ A reverse proxy is a server that is **positioned** in front of **web servers** to ensure the security of incoming data from external sources.
- ❖ A website may have several web servers **behind** reverse proxy.
- ❖ Takes **request** from **internet** and **forwards** it to one of the **web servers**.



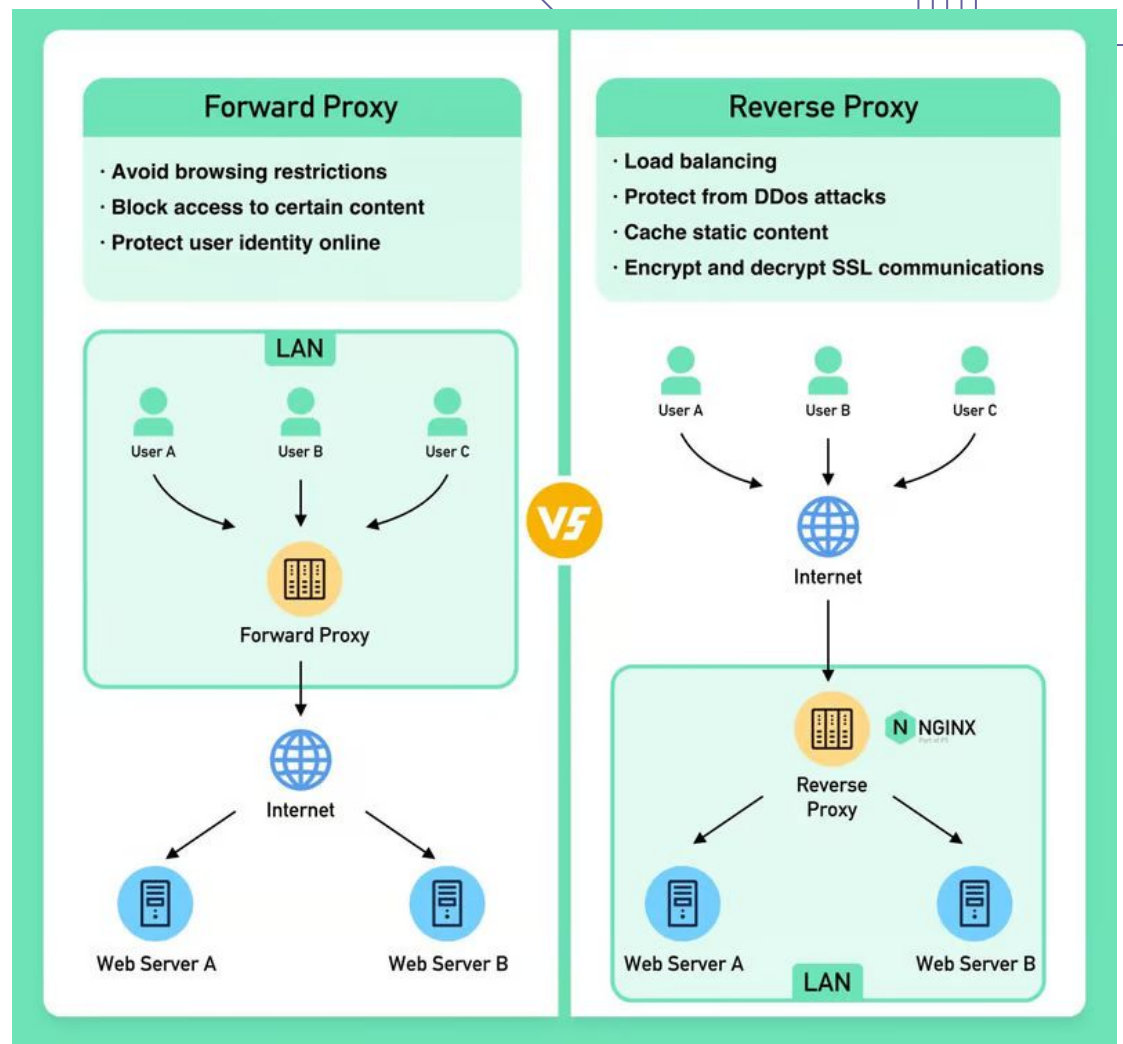
► Reverse Proxy

- ❖ A client **sends** the connection **request** to the **internet**.
- ❖ Instead of connecting the client **directly** to the web server, the **internet forwards** the client's connection request to a **reverse proxy**.
- ❖ The **reverse proxy** communicates with the **web server**.
- ❖ The web server **provides** the requested information to the **reverse proxy**.
- ❖ The response from the origin server is received by the reverse proxy and sent to the client.

How Does a Reverse Proxy Work?

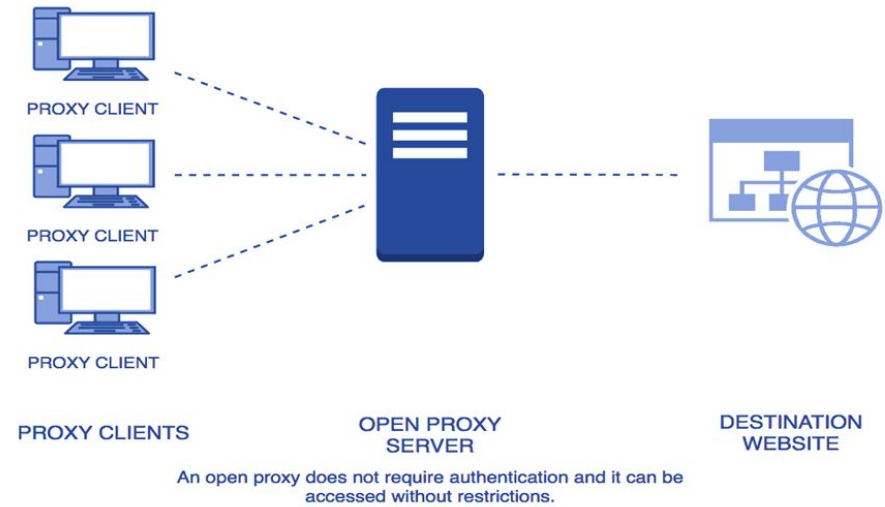


► Example:



▶ 3.Open Proxy

- ❖ Helps clients to **conceal** their IP address while browsing the web.
- ❖ Accessible by **any internet user**.
- ❖ Does not require authentication.
- ❖ They are, however, **free to use** and can come in **handy** when you need a **quick, one-time solution** for concealing your IP address.
- ❖ Public servers are generally considered **less safe** and **beneficial** than private proxies.



► Advantages of Proxy Server:

- ❖ It improves the security and enhances the privacy of the user.
- ❖ It hides the identity (IP address) of the user.
- ❖ It controls the traffic and prevents crashes.
- ❖ Also, saves bandwidth by caching files and compressing incoming traffic.
- ❖ Protect our network from malware.
- ❖ Allows access to the restricted content.

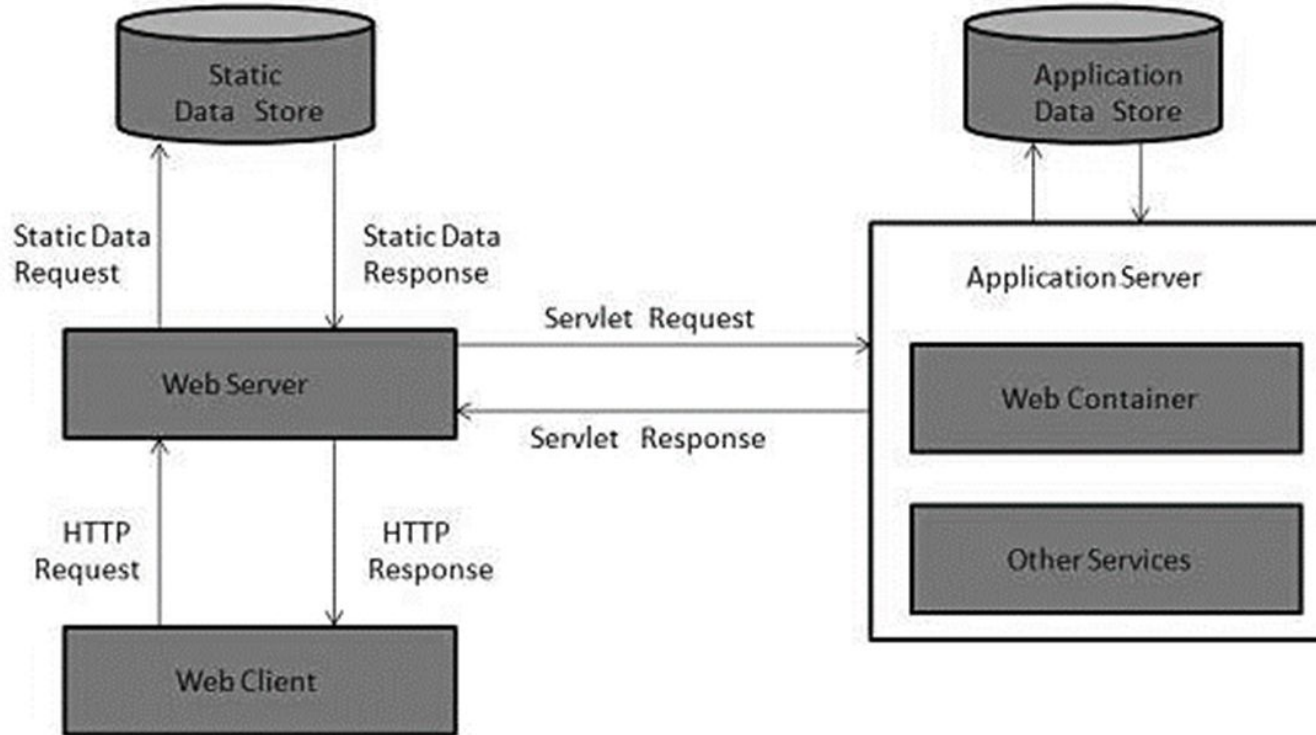
► Proxy Server vs VPN:

Basis of Comparison	Proxy Server	VPN
Encryption	It does not encrypt the traffic.	It encrypts the traffic.
Software	It does not have its own software.	It has its own software.
Speed	It is faster than VPN.	It is slower in comparison to the proxy server.
Cache	It uses cache.	It does not use cache.
IP Address	It hides the IP address but the proxy owner can see the client's IP address.	The IP address is completely hidden.
Connection	It is unstable.	It is seamless.
Pricing	It is mostly free.	It is chargeable.
Reliability	Its connection drops more frequently.	Its connection is more reliable.
Level of Working	It works on the application level.	It works on the operating system level.
Security	It is less secure.	It provides greater security.

► Web Application Server:

- ❖ It is a computer **where the web content** is stored.
- ❖ Used to **host websites** but there might consist other servers as well such as gaming.
- ❖ It is a software that **responds to request** for web resources.
- ❖ When client sends request for a web page, the web server **search for** the requested page, if not found it will send an **HTTP response: Error 404 not found**.

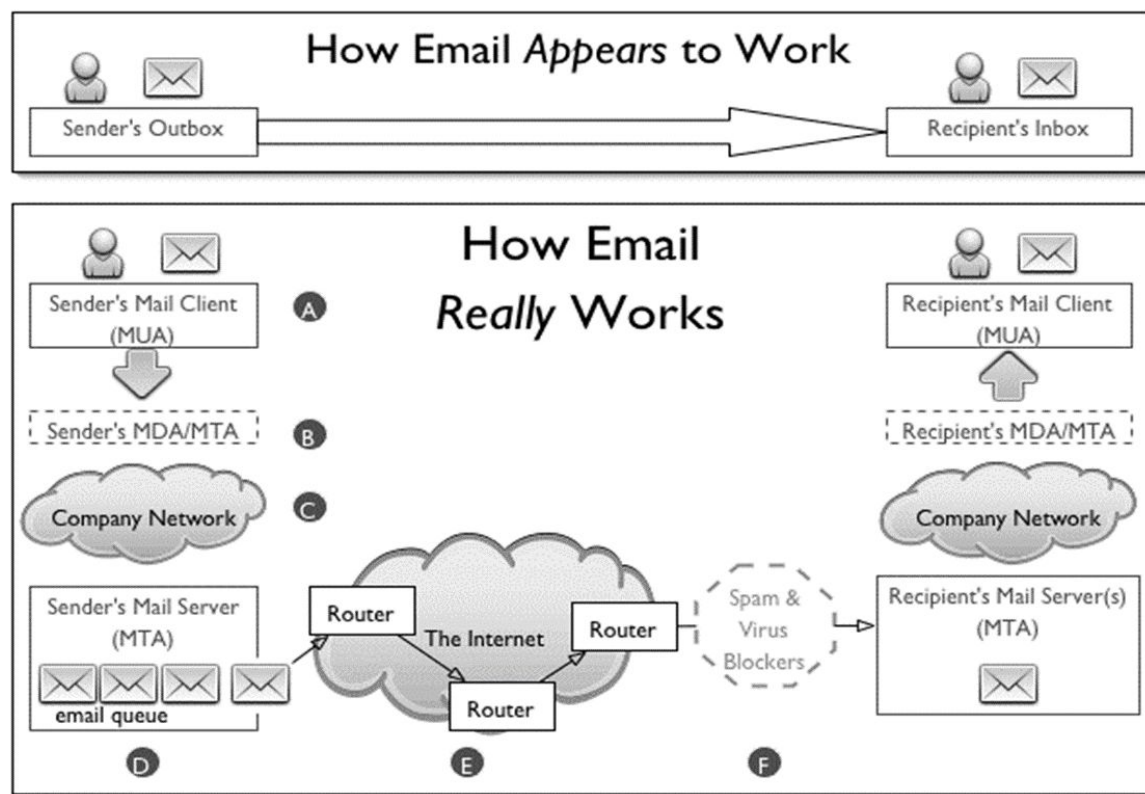
► Web Application Server:

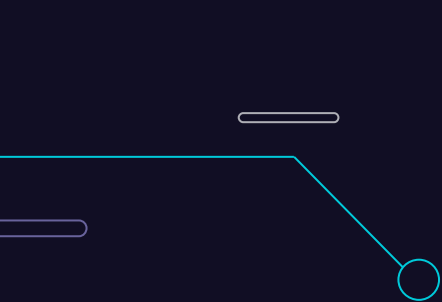


► Mail Application Server:

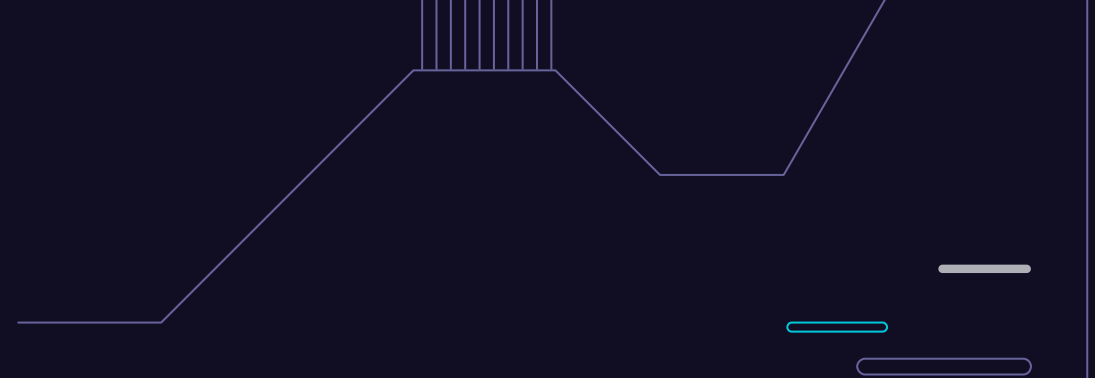
- ❖ Also known as mail transfer agent(MTA)
- ❖ It is an application that **receives** incoming e-mail from local users and **forwards** outgoing email for delivery.
- ❖ Mail servers have **dedicated computers** for such services.
- ❖ **Example:** Microsoft Exchange, Gmail
- ❖ When an email is sent, the clients email program **forwards** the message to **mail server**, which **in turn forwards** it either to **another mail server**.
- ❖ Mail servers can be broken into **two categories**:
 - **outgoing mail** servers known as SMTP servers
 - **incoming mail** servers know as POP3 servers

► Mail Application Server:





► Network Management: SNMP

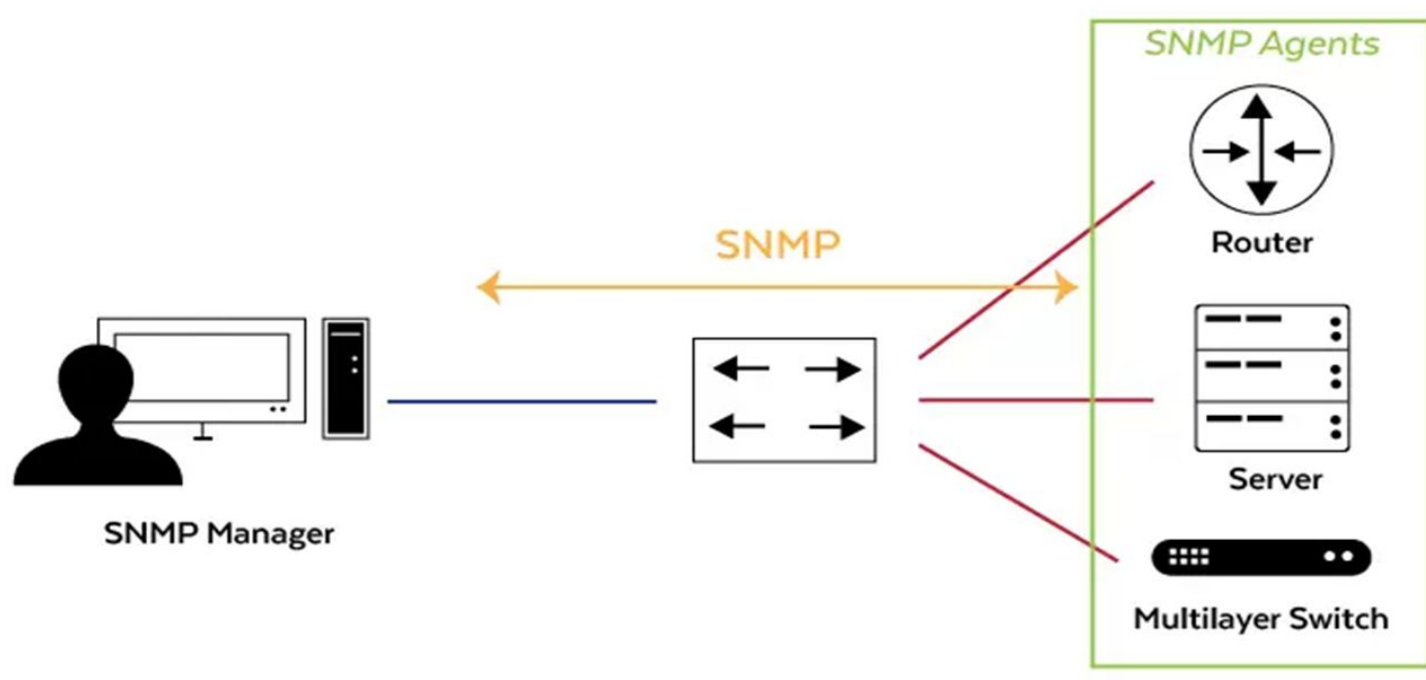


6.6

► SNMP:

- ❖ SNMP stands for **Simple Network Management Protocol**.
- ❖ It is an application layer protocol defined by Internet Architecture Board(**IAB**).
- ❖ It is a **part of** TCP/IP protocol suite.
- ❖ Protocol to **manage** and **monitor** network elements.
- ❖ Network element/devices come with **SNMP agent**.
- ❖ It is a set of protocols for network management and monitoring.
- ❖ Supported by **devices such as** routers, hubs, bridges, switches, servers, workstations, printers, etc.
- ❖ Helps to **obtain current condition** for proper and ongoing network administration.
- ❖ SNMP is an application layer protocol that uses **UDP port number** 161/162.

► SNMP:



► SNMP Provides Following Functionality:

- ❖ **Provide read/write ability:** example – helps to reset passwords remotely or re-configuring devices.
- ❖ Collect information about **bandwidth usage**.
- ❖ Collect **error reports** for troubleshooting.
- ❖ Email an **alert** when some problem occurs like low disk space, too much traffic on network or device fails.
- ❖ **Monitor** servers CPU and Memory use.

► Component of SNMP:

SNMP Consist of 4-components as:

❖ SNMP manager

- It is responsible to **communicate** with the **SNMP agent** implemented on network devices.
- It is typically **a computer** that is used to run **one or more** network management systems.

❖ Managed Devices

- It is a network element that requires some form of **monitoring** and **management**. Example: router, switch, workstation, etc.

❖ SNMP Agent

- It is a **program** that is packaged within the **network element/device**.
- Enabling the agent allows it to **collect the** management information database from device and makes it available to the SNMP manager, when it is **queried** for.

❖ Management Information Database (MIB)

► **Functions of SNMP Agent:**

- ❖ Collects management information about its environment
- ❖ Stores and retrieves management information.
- ❖ Signals event to manager.

► Management Information Base(MIB):

- ❖ Every SNMP agent **maintains** an **information database** describing the **managed device** parameters.
- ❖ SNMP manager uses this database to **request the agent** for specific information.
- ❖ It is a commonly **shared database** between SNMP agent and SNMP manager.
- ❖ MIB consists of information on resources that are **to be managed**. This information is organized **hierarchically**. It consists of objects instances which are essentially variables.
- ❖ A MIB, or collection of all the **objects** under management by the **manager**, is **unique to** each agent.

► Advantages/ Limitations:

Advantages of SNMP

- ❖ It is simple to implement.
- ❖ Agents are widely implemented.
- ❖ Agent level overhead is minimal.
- ❖ It is robust and extensible.
- ❖ Polling approach is good for LAN based managed object.
- ❖ It offers the best direct manager agent interface.
- ❖ SNMP meet a critical need.

Limitation of SNMP

- ❖ It is too simple and does not scale well.
- ❖ There is no object oriented data view.
- ❖ It has no standard control definition.
- ❖ It has many implementation specific (private MIB) extensions.
- ❖ It has high communication overhead due to polling.

► PYQs:

1. What are the features of the application layer? Why is DNS required? Explain about recursive, non-recursive, and iterative DNS queries. **2081**
2. What do you understand by DNS? Explain FTP and SFTP. **2080**
3. Explain DNS with reference to its hierarchy and records. **2080**
4. What is the function of proxy server? Explain about electronic mail. **2080**
5. Differentiate between IMAP and POP3. Explain SNMP. **2080**
6. Explain architecture of WWW. What is URL? **2079**
7. Why do we need a DNS system when we can directly use an IP address? what is domain name space? **2078**
8. What is the difference between User Agent (UA) and Mail Transfer Agent (MTA)?
9. In Client-Server Model what is the role of a Client program? What is the role of Server program?

▶ **THANKS!**

? **Do you have any questions?**

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